A STUDY TO EVALUATE THE EFFECTIVENESS OF A HOME BASED EXERCISE PROGRAM ON WORK RELATED UPPER LIMB MUSCULOSKELETAL DISORDERS AMONG TAILORS- AN EXPERIMENTAL STUDY

By

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In partial fulfillment of the requirements for the degree of

MASTER OF PHYSIOTHERAPY (MPT)

In

MUSCULOSKELETAL CONDITIONS

Under the guidance of

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I hereby declare that this dissertation/thesis entitled **"A study to evaluate the effectiveness of a home based exercise program on work related upper limb musculoskeletal disorders among tailors- an experimental study"** is a bonafide and genuine research work carried out by me under the guidance of Dr. Priyadarshnini Mishra (PT) PhD. Professor, ABSMARI there are no conflict of interest associated with this dissertation work.

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Date: **Signature**

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SL.NO	CONTENT	PAGE NO.
1	Introduction	1-10
2	Aim and Objectives	11-12
3	Hypothesis	13-14
4	Review of Literature	15-35
5	Methods	36-54
6	Data Analysis and Interpretation	55-113
7	Discussion	114-121
8	Limitation of the study	122-125
9	Future Recommendations	126-129
10	Conclusion	130-132
11	Reference	133-145
12	Annexure	146-151

INTRODUCTION

1.1 Introduction

One of the significant health issues in the workplace is work-related musculoskeletal disorder (WRMSDs), which is an ailment brought on by repetitive, physically strenuous and ergonomically challenging tasks. These disorders affect muscles, tendons, ligaments, joints, nerves and other soft tissues and supporting structures of the musculoskeletal system. They often present as pain, discomfort, stiffness, limitations of joint movement, muscle fatigue and functional impairments. Depending on the type of occupational exposure, symptoms can impact any of the various anatomical areas, such as the neck, shoulders, upper back, elbows, wrists and hands. MSK pain has a negative impact on physical function, quality of life, work performance and productivity and may lead to increased health care utilization and work absence [1-5].

Shoulder and upper limb disorders are a significant proportion of overall burden of musculoskeletal disorders. Shoulder pain is common in the adult population and can occur due to various musculoskeletal problems, such as rotator cuff disease, subacromial pain syndrome, adhesive capsulitis and other non-specific shoulder pain complaints [4-8]. These conditions may affect daily tasks that involve repetitive movements of the arm, upper-limb positioning, lifting and reaching. Loose range of motion and upper-limb function impairments can also impact a person's independence in performing activities of daily living (ADLs). Previous studies have shown that musculoskeletal symptoms affect physical function and quality of life, making early identification and proper management very important [6-10].

Tailoring is a job that requires sitting for long periods of time, repetitive work of the upper parts of the body, constant use of sewing machines and visual concentration. Tailors often engage in activities involving repetitive usage of the wrists and fingers, and coordinated movements of the shoulders and elbows, as well as prolonged static positioning of the upper back and neck. These occupational factors can involve a significant biomechanical stress, especially when precautions

of ergonomic arrangements, rest time and physical condition are not provided. Exposure to these work-related demands on the body can result in the development or aggravation of work-related musculoskeletal symptoms.

1.2 Work-Related Musculoskeletal Disorders Among Tailors

Tailors engage in some repetitive tasks during their working time, such as cutting fabric, stitching, measuring, handling clothes, or running sewing machines and finishing clothes. Many of these activities involve a prolonged forward bend of the neck, rounded positioning of the shoulders, repetitive patterns of wrist and hand movements and holding a static posture for prolonged periods. The continuous use of the sewing-machine might involve multiple foot-pedal pressing and releasing movements while the upper limbs are involved in focused and repetitive activities.

Static muscle contraction over long periods and repetitive actions may lead to an increase in mechanical load to the muscle and supporting structures. Lack of proper workstation height, improper adjustments of the chair, lack of back support, lack of light and limited rest periods can also cause discomfort in the workplace. Some studies have been conducted on various occupational groups, revealing that repetitive work, awkward body positions, sitting too long, poor work posture, and physically challenging jobs are important factors for the health of workers' musculoskeletal system [33-38, 54-59].

The nature of occupational exposure is different for different occupations but studies of industrial and health care workers have shown the association between physical work load and musculoskeletal symptoms. Other studies in the occupational field have focused on the effect of repetitive and physically demanding work [35-41] and Andersen et al. have studied musculoskeletal risk factors and long-term sickness absence among healthcare workers. Similarly, research with industrial technicians has shown the potential benefits of using particular resistance exercise to treat forearm pain and work disability [43].

Tailors can have discomfort in the upper extremity (shoulder, upper arm, elbow, forearm, wrist and hand). These structures might be required to provide support to the fabric during repeated handling or during long periods of small object manipulation or sewing activities.These structures may need to support the fabric during repeated handling, long periods of small object manipulation, or extended sewing activities. Forward head positioning and long periods of concentration on the work can also cause neck and upper-back pain.

WRMSDs aren't only a pain in the neck. Chronic pain can decrease work capacity, affect concentration, limit daily living and impact occupational obligations. Other consequences of musculoskeletal pain include decreased work productivity, sickness absence and decreased quality of life [33, 34, 38]. Thus, it is significant to recognize the occupational risk factors and to develop interventions to prevent and treat musculoskeletal problems among tailoring workers that are accessible to the workers.

1.3 Occupational Risk Factors and the Importance of Ergonomics

The development of most occupational musculoskeletal disorders is due to a combination of individual, physical and environmental factors. Musculoskeletal discomfort may be caused by repetitive movements, holding one position for too long, or inappropriate use of the workstation, having too short a rest period, or too much biomechanical loading.

Tailoring activities are especially of interest when considering prolonged sitting and repetitive upper-limb movements. Forward flexed position of the neck while focusing on stitching can add to a high level of sustained effort in the cervical and upper-back muscles. Likewise, repetitive shoulder and elbow movements, as well as sustained wrist and finger use can cause muscles to fatigue and get sore.

Research conducted in the field of occupational ergonomics highlights the significance of having the right conditions and setup for workstations and the importance of making workstation changes

to minimize MS stress. Koma et al. conducted a study on barriers and facilitators involved in the implementation of a workplace ergonomics programme, and found organizational and individual factors to be key in ensuring the continuity of workplace ergonomic practices [54]. It has also been laid out clearly from a systematic ergonomics research that appropriate work environment and working arrangements is important [56, 59].

While ergonomic modifications will decrease exposure to occupational risk factors, it may not be enough to overcome pre-existing muscle weakness, limited flexibility and functional limitations. Therefore, exercise interventions could represent another treatment strategy to enhance physical ability and alleviate musculoskeletal complaints.

1.4 Exercise-Based Management of Upper-Limb Musculoskeletal Disorders

Both exercise therapy and physiotherapy play important roles in treatment of musculoskeletal pain and in improving physical function. Exercise may consist of strengthening, stretching, mobility, postural correction or general physical conditioning. They are interventions for enhancing joint mobility, flexibility and functional activities and the performance of muscles.

Systematic reviews and clinical trials of exercise interventions have demonstrated evidence for different shoulder and upper limb conditions. Pieters et al. performed a literature review of conservative physical therapy for subacromial shoulder pain and found evidence to support the physiotherapy based management of subacromial shoulder pain [14]. In the same manner, Liu et al. conducted a systematic review and meta-analysis of the effectiveness of home-based exercise for treating non-specific shoulder pain [12]. The effectiveness of exercise and manual therapy for shoulder pain and function has been explored in other reviews [13, 20-25].

Certain strengthening activities could enhance the function of the muscles that stabilize the shoulder and move the upper limb. Ginn and Cohen explored exercise therapy to restore neuromuscular control of persons with shoulder pain [19]. Kuhn also discussed the therapeutic

exercises for rotator cuff impingement as a component of an evidence-based rehabilitation program [21].

Another form of exercise-based rehab is resistance training. Progressive resistance exercises can enhance muscle strength and endurance by progressively loading the desired muscle groups in controlled manner. Industrial technicians showed that certain resistance training led to a decrease in forearm pain and work disability [43]. Likewise, exercise interventions at work for the neck and shoulders have shown to benefit occupational musculoskeletal pain [44, 51].

Exercises of stretch and flexibility are designed to keep or enhance the extensibility of muscles and supporting soft tissues. These exercises can help to address movement restrictions caused by static positioning and/or repetitive tasks at work. Strengthening, flexibility and mobility exercises might then be a holistic treatment for upper-limb musculoskeletal symptoms.

1.5 Home-Based Exercise Programme

Home based exercise programmes are becoming a more readily available method of physiotherapy intervention, especially for people who might struggle to come to physiotherapy sessions regularly. These programs enable the individual to execute their workout routines in a setting where they are comfortable, using relatively less equipment and travel.

The efficacy of a home-based exercise program to enhance the function of the upper limb in community-dwelling older people was examined in a pragmatic randomized controlled trial (RCT) performed by Bates et al. Upper-limb function was evaluated using the Disabilities of the Arm, Shoulder and Hand (DASH) questionnaire and secondary outcomes included shoulder strength, shoulder range of motion and quality of life [1]. This study did not prove that there were any significant differences in the improvement of the upper-limb functions among the groups, but it did show that a structured upper-limb exercise programme could be learned and followed at home.

Liu et al. also reviewed the efficacy of home based exercise for shoulder complaints other than specific pain, and gave evidence relevant to the use of home exercise in shoulder rehabilitation [12]. Granviken and Vasseljen originally evaluated the effects of supervised and home based exercise interventions on the pain and disability of people with subacromial impingement and found similar results, after accounting for the dose of exercise training [32]. Christiansen and Hjort also explored the effects of group-based, individually supervised and home-based training in people with subacromial pain [31].

The effectiveness of home-based exercise can however be influenced by a number of factors which include the type of exercise, instructions, compliance, frequency, intensity and progression. Jordan et al. conducted a review of interventions to support exercise adherence in adults with chronic musculoskeletal pain, emphasising the need to keep individuals participating in prescribed exercise programmes [47].

A cluster randomized controlled trial was conducted to examine physical exercise in the workplace versus home setting for female health care workers (HCWs) by Jakobsen et al. The group in the workplace showed more improvement in musculoskeletal pain and muscle strength than the group at home and there were differences in adherence to the exercise program [2]. Findings indicate that participation and outcomes from an exercise programme may be influenced by the exercise setting and how the programme is delivered.

Regardless of these factors, a structured home based programme might be especially helpful for workers who do not have a lot of time or access to regular physiotherapy sessions. A program of simple exercises that can be performed at home using few (if any) pieces of equipment could be a practical way of addressing occupational musculoskeletal discomfort for tailoring workers.

1.6 Outcome Assessment of Musculoskeletal Disorders

The effectiveness of physiotherapeutical interventions should be evaluated using appropriate outcome assessment. Depending on the research aims and the anatomical area, the assessment of musculoskeletal disorders can be subjective and/or objective.

Pain intensity is an important subjective outcome since it is a reflection of how painful the pain is for the participant. There are various methods that are used to document the intensity of pain, such as the Visual Analogue Scale (VAS) and the Numerical Pain Rating Scale (NPRS). Pincus et al. raised some questions about the application of various visual analogue scale formats to collect pain and other clinical information [50].

Another significant outcome is functional disability, as pain can affect a person's ability to perform everyday tasks and work. The Disabilities of the Arm, Shoulder and Hand (DASH) questionnaire was developed to assess physical function and symptoms from upper extremity disorders [47, 48]. It is also has a short form (QuickDASH) used to assess upper-limb disability. These have been investigated for validity, reliability and responsiveness [47, 48, 50-52].

The range of motion (ROM) is an objective measure of the mobility of the joints, especially in shoulder, elbow, and wrist joint movements. An appropriate goniometer or inclinometer may be useful for measurement and to document changes after an intervention, to help identify movement restrictions.

Quality of life is also important as a consequence of musculoskeletal symptoms that persist, which could impact physical function, psychological well-being and participation in life. The association of musculoskeletal symptoms and physical function and quality of life has been explored in previous studies that focused on shoulder disorders [6, 9]. Hence, combining the assessment of pain, functional assessment and objective physical assessment can yield a more comprehensive picture of the impact of interventions.

1.7 Rationale of the Study

Tailors are an occupation, which often require repetitive use of the upper limb, long periods of sitting and demanding postures. These types of work exposures can lead to musculoskeletal symptoms and work-related functional limitations in the shoulders, arms, wrists and hands. Although occupational musculoskeletal health is relevant, accessible exercise interventions targeted to tailoring workers' needs need to be studied further.

Previous studies have shown that exercise-based interventions can be beneficial for the management of shoulder pain, increase muscle strength and prevent upper-limb dysfunction [12-14, 19-25]. The effectiveness of home-based exercise programmes has also been studied, as well as various methods of exercise delivery [1, 2, 31, 32]. However, results obtained from older adults, health workers and industrial workers cannot be extrapolated to tailors due to the different biomechanical demands and activities that they have.

A home-based structured exercise program that combines aerobic activity, resistance training, and flexibility exercises could be a feasible way to meet the physical demands of tailoring activities. This type of programme could aid in the improvement of the muscular conditioning, the maintenance of joint mobility and the reduction of perceived musculoskeletal discomfort.

The present study is thus aimed to assess the effectiveness of home based exercise programme for tailors in preventing work related UWMDs. The intervention will be carried out for eight weeks and the participants will be divided in the experimental and control group based on the methodology of the study. This will be assessed to determine the changes in pain, functional ability and range of motion.

The results could aid in the development of easy-to-use physiotherapy interventions for occupational groups that are subject to repetitive tasks of the upper limb. Moreover, the study could yield useful information for physiotherapists, occupational health professionals and tailoring

workers on the possibility of structured home-based exercise in the management of work-related musculoskeletal problems.

1.8 Statement of the Problem

Upper limb musculoskeletal disorders (MSDs) are an important occupational health issue for people who carry out repetitive and prolonged manual activities. Tailors often perform extended sitting positions, repetitive tasks and repetitive use of sewing machines, which may lead to pain, discomfort and functional impairment.

Exercise-based physiotherapy interventions have been shown to be effective in the management of a variety of musculoskeletal conditions, but the effectiveness of a structured home-based exercise programme in tailoring workers remains to be investigated.

The present study is, therefore, aimed at testing the effectiveness of home-based exercise programme for work related upper-limb musculoskeletal disorders among tailors.

1.9 Significance of the Study

Some findings from the study could be relevant to the question of whether home-based exercise can help to manage OUMS symptoms in tailors. Simple aerobic, strengthening and flexibility exercises can be used for a structured programme that may be an accessible intervention with minimal equipment.

The results could assist physiotherapists in creating an appropriate exercise advice for people that have been exposed to repetitive activities at work. The programme can also raise the awareness of tailoring workers on the importance of physical condition, movement and musculoskeletal health.

Finally, this study can be used to enhance the occupational well-being status and to promote the use of practical physiotherapy interventions of tailoring workers.

AIM AND OBJECTIVES

2.1 Aim of the Study

To evaluate the effectiveness of a structured home-based exercise programme on work-related upper-limb musculoskeletal disorders among tailors.

2.2 Objectives of the Study

Primary Objective

1. To evaluate the effectiveness of a home-based exercise programme in improving functional ability among tailors experiencing work-related musculoskeletal disorders.

Secondary Objectives

1. To assess changes in pain and discomfort following the home-based exercise programme.
2. To evaluate changes in the range of motion of affected upper-limb joints after the intervention.
3. To assess the prevalence and distribution of work-related musculoskeletal disorders among tailors.
4. To develop and implement a structured home-based exercise programme addressing the occupational musculoskeletal demands of tailoring.
5. To assess changes in functional mobility and work-related functional performance following the intervention.
6. To evaluate changes in quality of life following the exercise programme.

HYPOTHESIS

3.1 Null Hypothesis (H_0)

There will be no statistically significant effect of the structured home-based exercise programme on work-related musculoskeletal disorders, pain, functional ability and range of motion among tailors.

3.2 Alternative Hypothesis (H_1)

The structured home-based exercise programme will produce a statistically significant improvement in work-related musculoskeletal symptoms, pain, functional ability and range of motion among tailors.

REVIEW OF LITERATURE

4.1 INTRODUCTION

Among the occupational health issues related to repetitive activities, static postures, awkward working positions and sustained biomechanical loading are common WMSDs (work-related musculoskeletal disorders). These disorders could involve muscles, tendons, ligaments, joints and other supporting structures, causing pain, stiffness, muscular fatigue and loss of function. MS symptoms can limit daily living functions, limit productivity in the workplace and impact quality of life.

Tailoring is a job with repetitive upper-limb motions, often sitting, and needing to operate the sewing machine and maintain visual focus. These activities can cause significant musculoskeletal stress especially to the neck, shoulders, elbows, wrists and hands. Hence, it is crucial to know about possible risk factors in the occupation and to assess physiotherapy interventions where they may be appropriate, to ensure the musculoskeletal health of tailoring workers.

The following review discusses the prevalence and consequences of musculoskeletal disorders, exercises for prevention of occupational hazards, strengthening and flexibility exercises, home-based rehabilitation, ergonomic intervention, and functional assessment and adherence to exercise.

(A). SECTION I: LITERATURE RELATED TO THE PREVALENCE AND IMPACT OF MUSCULOSKELETAL DISORDERS

Urwin et al. (1998)

Urwin et al. explored the burden of musculoskeletal disorders in the community by looking at the prevalence of symptoms at different sites compared to each other and how this relates to social deprivation. Their study showed that generally MS symptoms are common and emphasised the need to think about the various anatomical areas when looking at the burden of MS. The authors also discussed the role of social and economic factors on osteo-muscular health, thus giving a good foundation to understand the occupational and community-related musculoskeletal issues [6].

van der Heijden (1999)

van der Heijden summarised the clinical features and treatment of shoulder diseases. Different types of shoulder conditions were discussed, including their clinical presentation, the need for proper assessment and management. The results emphasized the multi-faceted nature of shoulder disorders and the importance of taking into account the structural and functional aspects of the shoulder in planning rehabilitation. This review gives background and context to the musculoskeletal issues regarding the shoulder in workers who engage in repetitive upper-limb activities [4].

Luime et al. (2004)

Luime et al. carried out a systematic review to assess the prevalence and incidence of shoulder pain in the general population. The investigation showed that pain in the shoulder is an important musculoskeletal health problem affecting various population groups. The authors emphasized the significance of epidemiological evaluation to determine burden of shoulder disorders and implications for physical functioning [8].

Harkness et al. (2005)

Harkness et al. looked at whether a change in incidence of musculoskeletal pain was occurring over time by comparing two cross-sectional surveys that were collected in approximately 40-year intervals. They analyzed the trends in musculoskeletal pain and emphasized the relationship of MSDs as a public health issue. The study results were valuable in terms of information on the prevalence of musculoskeletal symptoms at the population level and in relation to suitable preventive measures [5].

Hill et al. (2010)

Hill et al. looked at the prevalence and factors associated with shoulder pain and stiffness in a population-based study, the North West Adelaide Health Study. They investigated the prevalence of shoulder-related symptoms and related factors in community members. The findings emphasized shoulder pain and stiffness as key musculoskeletal issues which can impact functional ability and overall health. This study furnishes background information for the measurement of shoulder mobility and symptoms in people who have been involved in repetitive upper limb activities [7].

Marks et al. (2019)

Marks et al. looked at the financial burden of shoulder pain on patients who underwent public orthopaedic referral in Australia. Their study not only identified some of the economic costs of shoulder disorders but also illustrated that musculoskeletal conditions can have impacts beyond physical complaints. The findings highlighted the need to take into account the health care utilization and other socio-economic impacts of shoulder diseases [10].

Lucas et al. (2022)

A systematic review was performed by Lucas et al. to review the prevalence and incidence of shoulder pain all over the world. Their study gave epidemiological data on the prevalence of shoulder symptoms among various populations and geographical areas. The results confirmed the significance of shoulder pain as a musculoskeletal health problem and the necessity of the proper preventive and therapeutic measures [9].

(B). SECTION II: LITERATURE RELATED TO OCCUPATIONAL RISK FACTORS AND WORK-RELATED MUSCULOSKELETAL DISORDERS

Skotte et al. (2002)

Skotte et al. performed a dynamic 3D biomechanical analysis of the loading of the lower back in various patient handling tasks. They studied the biomechanical challenges of the different occupational activities and showed the need to consider physical loading during work-related movements. The investigation was centered on patient-handling activities, but the results of this investigation has been useful in understanding musculoskeletal stress as a factor of repetitive and physically demanding task in the workplace [37].

Lund et al. (2006)

Lund et al. conducted a prospective cohort study of 5,357 workers in Denmark to examine risk factors for long-term sickness absence in the physical work-environment. Their study identified the link between the physical environment at work and health at work. The results highlighted the need to take into account work-related risk factors when assessing musculoskeletal issues and designing preventive occupational health strategies [36].

Bigos et al. (2009)

After performing a systematic literature search of high-quality controlled trials in adults aged 18-45 years regarding the prevention of back problems, Bigos et al. concluded that there was a lack of evidence-based information to support the prevention of back problems in the working-age population. They looked at ways to prevent occupational back problems. The results indicated evidence of the importance of preventive interventions and the necessity of the development of the programmes corresponding to the research and development of adequate programmes to preserve the musculoskeletal health of the working population [46].

Holtermann et al. (2010)

Holtermann et al. presented the background, design and conceptual model of FINALE, a worksite project for the prevention of physical deterioration in workers under high physical work demands. They focused their study on the need to design and implement preventive programmes in the

workplace for physically demanding groups. The study suggested a framework for understanding the contribution of structured physical interventions in maintaining the physical capacity of workers [35].

McDonald et al. (2011)

McDonald et al. conducted a study on the impact of back pain, arthritis and fibromyalgia pain on workers' quality of life and their productivity at work. They found that musculoskeletal pain could affect the occupational performance and overall health of people. Results also pointed to the wider impacts of chronic musculoskeletal problems and to the need to include functional limitations in occupational health interventions related to pain [33].

Andersen et al. (2012)

The Andersen et al. study investigated the point of musculoskeletal pain intensity that indicates a higher risk of long-term sickness absence in female health care workers in eldercare. They found that the level of pain intensity was related to occupational function. The results showed that early detection of musculoskeletal symptoms can have a positive impact on the consequences to work participation [34].

Andersen et al. (2012)

The Andersen et al study was a prospective cohort study which focused on musculoskeletal risk factors for sickness absence over a long period of time among health care workers in an elder care setting. They found that physical exposure in the workplace is significant in the occurrence of work-related health issues. The study offers relevant evidence of the association of musculoskeletal risk factors with occupational functioning [38].

Andersen et al. (2014)

To examine the association between patient transfers, use of assistive devices and occupational back injury within health care settings, Andersen et al. conducted a prospective cohort study. They studied how physical work injuries affected by occupational activities can impact their research. The results emphasised the role of correct working methods and the usage of assistive means, to minimise the occupational biomechanical loading [40].

Burdorf et al. (2013)

Burdorf et al. studied the effect of lifting devices on low-back pain and musculoskeletal injury claims of nurses. They found that occupational adaptations to prevent physical loading and musculoskeletal injury are important. The findings give relevant background information on workplace intervention to prevent WMSD [41].

Miranda et al. (2020)

Miranda et al. explored working conditions and their effects on the health of nursing professionals in the COVID-19 pandemic. They identified the link between work demands, working conditions and employee health. The study was targeted on nursing professionals but the outcomes have been utilized to give a wider occupational health context on the significance of acceptable working circumstances and preventive measures [63].

(C). SECTION III: LITERATURE RELATED TO EXERCISE-BASED MANAGEMENT OF MUSCULOSKELETAL DISORDERS

Andersen et al. (2008)

The function of chronically painful muscles after physical training was the subject of a randomized controlled study by Andersen et al. They studied the potential benefits of a structured physical train in people with persistent symptoms of muscular pain. This study confirmed the exercise-based intervention effect on the muscular function and evidence was given for the inclusion of therapeutic exercise in musculoskeletal rehabilitation programmes [45].

Kuhn (2009)

Kuhn performed a systematic review that explored exercise for the management of rotator-cuff impingement and formulated a synthesized evidence-based exercise-based rehabilitation plan. The application of therapeutic exercises for shoulder related symptoms and for improvement of functional capacity was addressed in the review. The results emphasized the need to provide structured rehabilitation, including exercises that are both strengthening and mobility exercises [21].

Ginn and Cohen (2005)

Ginn and Cohen performed a randomized comparative clinical trial (RCT) that looked at exercise therapy to restore neuromuscular control for shoulder pain. Their studies revealed the value of targeted exercise interventions to increase the function of the shoulder and re-establish the proper coordination of the muscles. Therapeutic exercise can be employed in people with shoulder musculoskeletal symptoms and symptoms are relevant to the use of therapeutic exercise [19].

Gebremariam et al. (2014)

Gebremariam et al. studied the efficacy of physiotherapy and manual therapy for SIS. They reviewed the conservative treatment options for shoulder problems, and they discussed the evidence for the use of physiotherapy in shoulder symptoms. The results helped to understand rehabilitation strategies that can improve shoulder function and discomfort [20].

van den Dolder et al. (2014)

In a systematic review and meta-analysis (SRM), van den Dolder et al. analyzed the effect of soft-tissue massage and exercise for the treatment of non-specific shoulder pain. The role of conservative treatment in shoulder symptoms was evaluated by their research. The study had evidence about the use of exercise and soft-tissue techniques in shoulder rehabilitation [13].

Abdulla et al. (2015)

Abdulla et al. performed a systematic review of the effects of exercise for subacromial impingement syndrome (SAIS) and other soft tissue injuries of the shoulder. In their investigation, they assessed conservative rehabilitation strategies and emphasized the need for exercise-based management for patients with shoulder-related musculoskeletal issues [25].

Steuri et al. (2017)

Steuri et al. performed a systematic review and meta-analysis of RCTs that compared conservative treatment modalities (exercise, manual therapy, medical management) for adults with shoulder impingement. Their study compared the effectiveness of various treatment options, and supported conservative management for musculoskeletal disorders of the shoulder [22].

Liaghat et al. (2021)

Liaghat et al. performed a systematic review and meta-analysis of supervised training vs. no training or self-training in patients with subacromial pain syndrome. They looked at the work of supervision in exercise-based rehabilitation. The results yielded valuable evidence about the impact of the mode of exercise delivery and how supervision might affect outcomes of the rehabilitation process [23].

Silveira et al. (2024)

The study by Silveira et al. was a network meta-analysis of the efficacy of shoulder-specific exercise therapy to reduce chronic shoulder pain. They showed that various exercise methods were comparable and that focused rehabilitation of the shoulder is important. Findings concur with the use of structured shoulder-specific exercise intervention in persistent shoulder related symptoms [24].

Zhong et al. (2024)

To determine the efficacy of scapular stabilization exercises in patients with subacromial pain syndrome, the effects of scapular stabilization exercises on subacromial pain syndrome were evaluated using a systematic review and meta-analysis of randomized controlled trials (RCTs). They studied the effect of scapular stabilization in the shoulder rehabilitation. The results justifies the incorporation of targeted stabilization exercises in rehabilitation programs of people with shoulder pain [26].

Dube et al. (2024)

Dube et al. performed a scoping review of exercise programmes for the treatment of shoulder pain associated with rotator-cuff problems. They reviewed various exercise programme characteristics such as frequency, intensity, time and type of therapeutic exercises. The study was helpful in appreciating prescription of exercise and the design of structured rehabilitation programmes for shoulder related musculoskeletal conditions [27].

(D). SECTION IV: Literature relating to strengthening and resistance exercises

Andersen et al. (2008)

In the study by Andersen et al., the authors investigated the outcomes of two different forms of physical activity in chronic pain of the neck muscles. Their research focused on exercise interventions for chronic neck pain and muscle performance. The results were used to support the use of physical exercises that were well matched to those who presented with chronic neck-related musculoskeletal problems [53].

Andersen et al. (2011)

To evaluate the effectiveness of a brief period of progressive resistance training administered on a daily basis for frequent neck and shoulder pain, Andersen et al conducted a randomized controlled trial (RCT). They found that structured and progressive resistance exercises are an important

component in the management of occupational musculoskeletal symptoms. The study also featured the benefits of comparatively brief exercise periods for the health of the neck and shoulders [51].

Andersen et al. (2012)

The study by Andersen et al. was a cluster randomized controlled trial for the effect of specific forearm resistance training on forearm pain and work disability in industrial technicians. They were mainly interested in repetitive loads in the field of occupational activities and used muscles that were exposed to work-related loading in their research. The results corroborated the feasibility of using targeted resistance training to decrease occupational forearm pain and enhance occupational functioning [43].

Zebis et al. (2011)

Zebis et al. performed a randomized controlled trial of neck and shoulder exercises for occupational pain relief of industrial workers. Their studies identified that targeted exercise programmes are important for workers who participate in repetitive upper-body activities. The study supported the application of specific exercises of the neck and shoulder for occupational musculoskeletal management [44].

Kim et al. (2021)

Kim et al. undertook a scoping review to explore the (physiotherapy) use of elastic-band exercises in older adults for enhancing shoulder function. Their analysis focused on the utilization of elastic resistance in the field of shoulder rehabilitation and its possible benefits to increase the function of muscles. The results were used to establish a basis for the use of resistance band exercises in rehabilitation programs [28].

Li et al. (2024)

The study was a mixed-methods systematic review by Li et al., which investigated the impact of elastic-band resistance training in the physical and mental well-being of elderly people. They investigated the possibility of benefits from elastic resistance training on physical health and functional capacity. The research offered more evidence of the use of resistance-band exercise in structured rehabilitation programmes [29].

(E). SECTION V: LITERATURE RELATED TO HOME-BASED EXERCISE PROGRAMMES

Granviken and Vasseljen (2015)

Granviken and Vasseljen performed a randomized trial of individuals with subacromial impingement (SAI) comparing home exercises with supervised exercises. They studied to see if exercise done outside of a clinic setting could have similar clinical results as supervised exercise rehabilitation. The results showed no difference between the effectiveness of home-based and supervised exercise programmes when the amount of exercise delivered was the same. The study revealed home-based rehabilitation as a potential accessible alternative to supervised exercise [32].

Liu et al. (2022)

Liu et al. did a systematic review and meta-analysis on the efficacy of home exercise on non-specific shoulder pain. Their study assessed the effectiveness of home exercise to control symptoms associated with shoulder problems and offered evidence that can be relevant to rehabilitation programs that can be carried out in the home environment. The study emphasized the value of using home-based exercise as an efficient method in shoulder rehabilitation [12].

Christiansen and Hjort (2021)

Christiansen and Hjort randomly allocated people with subacromial pain to a group exercise, individually supervised exercise and home-based exercise program. Different ways of delivering

exercise were studied for their clinical effects and cost-effectiveness. The results showed that the three methods had comparable clinical outcomes, and emphasized the possibility of home rehabilitation being cost-effective [31].

Bates et al. (2026)

A pragmatic randomized controlled trial was performed by Bates et al. to assess the effectiveness of a home-based exercise programme on upper limb function in an older community population. The study involved 617 participants in upper and lower limb exercise groups. The shoulder and upper limb strengthening and mobility exercises were integrated into the upper limb programme. Participants were provided with resistance bands, light weights and illustrated instructions for exercises. Disabilities of the Arm, Shoulder and Hand (DASH) questionnaire was used to assess upper-limb function. There was no clinically important or statistically significant difference in how well patients functioned in the upper limb between the two groups at 12 months. However, these findings have given key insights into the feasibility, acceptability and delivery of the home-based exercise programmes for older adults [1].

O'Keeffe et al. (2017)

To determine if there was a difference in effectiveness between group-based and individual physiotherapy exercise programmes for musculoskeletal conditions, O'Keeffe et al. carried out a systematic review and meta-analysis. Their research involved a comparison of the various methods used to deliver exercise, and evidence was presented about the effectiveness of alternative exercise settings for rehabilitation. Accessibility, supervision and programme organisation were identified as key factors to consider when choosing an appropriate method of delivering exercise in this study [30].

(F). A review of the literature related to workplace-based and home-based exercise intervention is provided in this section.

Jakobsen et al. (2014)

A study protocol for a single-blinded cluster randomised controlled trial that included healthcare workers and compared the two intervention groups was published by Jakobsen et al. The proposed intervention was a comparison of the effect of various exercise environments on musculoskeletal pain. The protocol emphasized the need to compare the environment of exercise delivery, as well as the outcomes of participation and intervention across various occupational groups [49].

The last reference cited in the provided information is from Jakobsen et al. (Not specified).

A cluster randomized controlled trial was conducted by Jakobsen et al. to examine the impact of PE at the workplace compared to PE at home on musculoskeletal pain in healthcare workers. The study evaluated the differences in participation in exercise and musculoskeletal outcomes between the two delivery settings. The results from the provided reference emphasize the significance of exercise setting and participation in assessing occupational rehabilitation programmes [2].

Beneka et al. (2024)

This specific exercise programme was the topic of a randomized controlled trial by Beneka et al. who wanted to investigate its effectiveness in reducing work-related neck and upper-back pain and enhancing mood state in various occupational groups. The study examined the effectiveness of structured exercise as an intervention for occupational MSD. The results showed that focused physical activity interventions can be beneficial in decreasing work related discomfort and enhancing overall well-being of participants [3].

(G). SECTION VII: Literature that is related to exercise adherence and participation.

Jordan et al. (2010)

Jordan et al. performed a Cochrane review of interventions aimed at enhancing adherence to exercise in adults with chronic musculoskeletal pain. Their study emphasized the need to establish

impediments to exercise participation and come up with measures to promote frequent compliance. The review presented a premise on the basis of which behavioural and practical aspects that can determine the success of home-based rehabilitation programmes can be understood [47].

Karlsson et al. (2014)

Karlsson et al. compared pain and functional outcomes of two home exercise programmes in women with chronic neck pain. In their clinical trial, a specific focus was on programme completers and responders. The study also emphasized the need to look at adherence and response of participants in the interpretation of the results of home based exercise interventions [48].

Bates et al. (2026)

Bates et al. assessed the acceptability and feasible implementation of a home-based upper-limb exercise programme in older adults in the community. The programme included systematic exercise guidelines and a demonstrative material. Their results were helpful in terms of the intention of the participants to keep exercising and the obstacles that could influence their long-term adherence to exercise. The research paper emphasized the need to offer clear guidelines and proper support in an attempt to promote the compliance with home-based exercise programmes [1].

(H). SECTION VIII: Literature in the subject of ERGonomics and Workplace Modifications.

MacEwen et al. (2015)

MacEwen et al. carried out a systematic review of standing and treadmill desk in the workplace. Their study tested the workplace interventions that were aimed at changing the sedentary behaviour and enhancing the patterns of physical activity. The review has given background on the significance of changing the long sitting working conditions and the need to consider workplace based interventions to enhance physical health [60].

Janneck et al. (2018)

Janneck et al. examined the ergonomic aspects of mobile work space design. Their study revealed the necessity of optimizing working conditions to the personal physical needs and integrating the ergonomics principles into the work space organization. The paper was valuable as a background to the importance of proper workplace set-ups in alleviating workplace physical strain [57].

Koma et al. (2019)

Koma et al. examined the obstacles and facilitators that were related to the implementation of office ergonomics programme in a South African research organization. Their research noted the significance of organizational and individual involvement in ergonomic interventions. The results highlighted that effective ergonomic programmes need proper planning, participation of the employees and organizational support [54].

de Macedo et al. (2020)

A systematic review on ergonomics and telework was carried out by de Macedo et al. The study investigated the connection between working conditions and ergonomic practices in home working conditions. The results indicated the significance of proper workstation ergonomics, as well as occupational environmental interventions to sustain musculoskeletal health [56].

Buomprisco et al. (2021)

Buomprisco et al. analyzed health and teleworking, commenting on the new occupational health issues in the wake of the COVID-19 pandemic. Their study emphasized the need to focus on working conditions, physical activity and ergonomic set-ups when assessing the occupational health of remote working settings [55].

Wutschert et al. (2022)

The study by Wutschert et al. is a systematic review of the working conditions and occupational health in home-office settings. They focused their study on the linkage between work-at-home and occupational health outcomes. The results gave good background information on the significance of ergonomic work environments and proper body exercise during work at long durations [59].

Fischer et al. (2023)

The study by Fischer et al. explored the role of both personal and organizational resources in the outcomes of home-based work during the COVID-19 pandemic among the employees of the government. Their study investigated how workplace resources help employees to be resilient and functional at work. The results also added to the background about the value of organizational support and appropriate working conditions in sustaining the wellness of employees [58].

(I). SECTION IX: MUSCULOSKELETAL Assessment and functional outcomes relating to literature.

Kuorinka et al. (1987)

The Standardised Nordic Questionnaires are the instruments developed by Kuorinka et al. to analyse the musculoskeletal symptoms. The questionnaire offers a systematic way of determining symptoms in various parts of the anatomy and is very applicable to occupational musculoskeletal studies. The emergence of this tool gave a valuable baseline to the systematic recording of musculoskeletal complaints and their distribution among occupational groups [52].

Pincus et al. (2008)

Pincus et al. reviewed various forms of visual analogue scale to measure pain and other clinical data. Their study emphasized the significance of proper measurement formats in clinical measurement. The investigation included informative information about methodology of using visual analogue scales to record the pain levels and other patient-reported outcomes [50].

Davis et al. (2023)

The research by Davis et al. aimed to estimate the relationship between the shoulder dysfunction and mobility limitation in older adults involved in the Baltimore Longitudinal Study of Aging. Their study brought forward the correlation between functional issues associated with the shoulder and overall mobility. The results highlighted the significance of measuring upper-limb functions and musculoskeletal symptoms [18].

Rickert et al. (2020)

Rickert et al. implemented a preliminary study based on accelerometers, to investigate the effect of shoulder immobilization on daily physical activity. Their research was an indication of the connection between physical activity and restriction of shoulder movements. The results emphasized the relevance of the correct movement and mobility of the shoulder [16].

Coleman and Clifft (2010)

Coleman and Clifft investigated the impact of shoulder immobilization on balance in older adults living in the community. Their research examined the links between upper-limb movement restriction and functional balance. The results also gave more background information on the extended functional implications of shoulder conditions [17].

(J). SECTION X: LITERature related to work place health and occupation physical activity.

Rathonyi et al. (2021)

Rathonyi et al. examined the physical activity and sedentary behaviour of workers during the COVID-19 pandemic. The study has investigated the change in occupational and daily physical activity patterns. The results indicated the need to observe the occurrence of sedentary behaviour and ensure that the levels of physical activity are adequate to promote occupational health [61].

Kgakge et al. (2021)

Kgakge et al. came up with a scoping review protocol to explore the distribution of work-related musculoskeletal disorders among nurses in sub-Saharan Africa. In their study, they aimed to examine the burden of musculoskeletal disorders among occupational groups in a healthcare workforce and emphasize the necessity of systematic evidence on the distribution of such disorders. Though the study did not use tailoring workers, the relevant regional occupational health context was still found in the study [64].

Munabi et al. (2014)

In a cross-sectional study carried out in Uganda, Munabi et al. examined risk factors of musculoskeletal disorders among nurse professionals working in low-resource areas. They researched occupational variables that related to musculoskeletal issues and emphasized the need to measure physical working conditions in resource-constrained settings. The results give valuable context on occupational musculoskeletal health in the developing-country context [65].

Simoens et al. (2005)

A study by Simoens et al. explored the issue of nursing shortage in OECD countries using an OECD Health Working Paper. The report is more focused on the workforce shortages; however, it also offers more occupational and health care workforce context. Its presence is only important as a background to the study of obstacles to occupational workforce but not as a direct indicator of musculoskeletal disorders [39].

Liu et al. (2022)

The COVID-19 pandemic caused Liu et al. to give an overview of open innovation in the healthcare sector. Their study mentioned the innovation in healthcare and the reaction to crisis conditions. It does not explicitly examine musculoskeletal disorders or physiotherapy exercise, but rather it offered more contextual information about innovation in healthcare service delivery [62].

SECTION XI: Research Gap.

The literature reviewed illustrates that musculoskeletal disorders are a significant occupational health issue in relation to repetitive tasks, long-term stationary postures and constant biomechanical loads. Past research has explored shoulder pain, neck pain, physical exposure in the workplace and the implications of musculoskeletal symptoms on the functional capacity, quality of life and productivity.

Some of the studies have indicated the possibilities of resistance, strengthening, stabilization and mobility exercises to manage musculoskeletal conditions. The studies by Andersen et al. (2011, 2012), Zebis et al. (2011), Ginn and Cohen (2005), and Zhong et al. (2024) suggest that there are evidence-based studies that support targeted exercise interventions to address the symptoms of the neck, shoulders, and upper limbs. Likewise, other researchers, such as Granviken and Vasseljen (2015), Christiansen and Hjort (2021), Liu et al. (2022), and Bates et al. (2026) have also analyzed home-based exercise strategies among various populations.

Nevertheless, the majority of existing intervention research has targeted healthcare professionals, factory workers, elderly persons and people with special shoulder issues. Such results cannot be directly applied to the tailoring of workers due to the fact that tailoring is associated with certain job requirements, such as frequent sewing actions, constant sitting and sitting in one place, having an upper limb, upper-limb sitting and constant use of the shoulder, elbow, wrist and hand joints.

Moreover, there is also a lack of evidence on the effectiveness of a structured home-based exercise programme that is specifically tailored to workers with the upper-limb musculoskeletal symptoms. Several outcomes such as pain, upper-limb functional limitations, range of motion and work-related impact on quality of life need to be studied in tandem as well.

Thus, the current research paper will fill this gap in the research by assessing an eight-week structured home-based exercise programme in tailoring workers. The research investigates the

changes in pain and musculoskeletal symptoms, upper-limb functional limitations, work-related impact and quality of life, and upper-limb range of motion after intervention.

Findings can help in the formation of accessible and job-specific physiotherapy intervention to customize workers and other job populations with repetitive upper-limb activities.

METHODS

The methodology outlines the research design, study setting, recruitment of participants, sampling procedure, sample size determination, outcome measures, variables, operational definitions, measurement tools and data collection procedure and experimental intervention to validate the effectiveness of the home-based exercise programme on the upper-limb musculoskeletal disorder of tailors.

5.1 Study Design

This study is scheduled to be an experimental study with a randomized controlled trial (RCT) design to test the effectiveness of structured home-based exercise programme in reducing work-related upper-limb musculoskeletal disorders among tailors.

The design will involve a comparison of results for an experimental group in the home-based exercise programme and a control group. Pre- and post-measures will be used to evaluate responses to the intervention to identify change in pain, functional ability, range of motion and other outcome measures specified by the participants.

There will be two groups in the study:

Experimental group: Participants will receive the structured 8 week home-based exercise programme.

Control group: these patients will be provided with the control condition outlined in the approved study protocol.

Assessment of baseline and post intervention will be conducted. The intervention should last for eight weeks; an introductory phase from week 1–4 and an improvement phase from week 5–8.

5.2 Study Setting

The study will be carried out at Skill Development Institute, Bhubaneswar, Odisha, India. The institute will also be used for recruiting and initial assessment for qualification of tailoring workers.

Individuals will be approached and screened based on the set inclusion/exclusion criteria. The home-based exercise programme will be carried out in the participants' home environment following appropriate instructions from the investigator.

Permission to access and conduct recruitment of the participants will be taken from the relevant government authorities of Skill Development Institute, Bhubaneswar before beginning the recruitment of participants.

5.3: Participant Recruitment and Sampling Method

5.3.1 Study Population

The study will be carried out on tailors (Male and Female) engaged in tailoring by machine. The participants will be collected from the Skill Development Institute Bhubaneswar based on the eligibility criteria mentioned in the research proposal.

5.3.2 Sampling Method

Eligible participants will be recruited using a convenience sampling method. Convenience sampling is one of the non-probability sampling techniques that involves selecting participants who are easy to access, available and willing to participate.

Potential participants will be approached at the study setting and provided with information about the purpose, procedures, length and requirements of the research. Persons who meet the criteria will be asked to volunteer.

Eligible participants will be enrolled after screening and following written informed consent. Randomization procedure as outlined in the approved RCT protocol will be followed for group allocation. The recruitment process and the allocation to groups are two distinct processes.Recruitment and group allocation are two different processes (convenience sampling).

5.3.3 Inclusion Criteria

- People aged from 20 to 45 years.
- The Tailors (both men and women).
- Persons engaged in tailoring using machines.
- At least one year's professional tailoring experience, tailors.

5.3.4 Exclusion Criteria

- Trauma or surgery in last six months.
- Persons who have a musculoskeletal or neuromuscular disorder as described in the original study proposal.

5.3.5 Participant Screening

All interested parties will be screened for eligibility first. Screening will consist of determining age, confirming tailoring experience and machine-based tailoring activity, relevant history of trauma/surgery, screening for known musculoskeletal and neuromuscular conditions, and consent to participate.

A participant information sheet will be provided to eligible participants outlining the research. Informed consent in written form will be obtained prior to enrolment and data collection.

5.4 Sample Size Calculation

The size of the sample will be calculated prior to participant recruitment so that the study will have sufficient statistical power to demonstrate a meaningful difference in outcomes between the experimental and control groups. The size of the samples is stated as 'yet to be calculated' in the original proposal for IEC and will not be specified until the statistical assumptions and calculations have been made.

5.4.1 Parameters Required for Sample Size Calculation

Parameter	Description
Study design	Randomized controlled trial
Primary outcome	To be confirmed in the finalized protocol
Significance level (α)	0.05, if approved
Statistical power	To be specified
Effect size	To be obtained from a relevant previous study or pilot data
Allocation ratio	To be specified
Expected dropout rate	To be estimated and justified

5.4.2 Sample Size Formula

For comparing the mean outcome between two independent groups, the following formula may be considered:

$$\mathbf{n = 2\,(Z\alpha/2 + Z\beta)^2\,\sigma^2 / \Delta^2}$$

Where:

n = required sample size per group.

Zα/2 = standard normal value corresponding to the significance level.

Zβ = standard normal value corresponding to the desired statistical power.

σ = estimated standard deviation of the primary outcome.

Δ = expected difference between the groups.

The calculated sample size may subsequently be adjusted for anticipated participant dropout using the formula:

Adjusted sample size = n / (1 − d)

Where d represents the anticipated dropout proportion. The final sample size will be divided between the experimental and control groups according to the approved allocation ratio. A numerical sample size must be calculated using the selected primary outcome, a suitable previous study and documented statistical assumptions. An arbitrary number should not be inserted.

5.5 Outcome Measures, Variables and Operational Definitions

Structured home-based exercise programme will be assessed for its effectiveness using functional ability, pain and discomfort, range of motion and quality of life. The exact instruments and scoring process should align with the protocol approved.

5.5.1 Independent Variable

The independent variable will be structured eight-week home-based exercise programme for tailors with work-related musculoskeletal complaints. The programme will include aerobic, resistance and flexibility activities which will be provided in a booklet or via video.

5.5.2 Dependent Variables

- Pain and discomfort.
- Functional ability.
- Limit of motion of affected joints.
- Quality of life.
- Functional mobility and functional performance at work.

5.5.3 Operational Definitions

Variable	Operational definition	Assessment
Work-related musculoskeletal disorders	Musculoskeletal symptoms associated with occupational activities	Participant screening and symptom assessment

	involving repetitive movements and sustained postures.	
Pain intensity	Subjective intensity of musculoskeletal pain experienced by the participant.	Standardized pain assessment tool to be finalized
Functional ability	Ability to perform activities involving the upper limbs without significant pain or limitation.	Functional disability questionnaire
Range of motion	Available movement of affected joints in different anatomical planes.	Inclinometer
Quality of life	Participant's perceived physical and general well-being.	Standardized quality-of-life questionnaire
Functional mobility	Ability to perform relevant functional movements and activities.	Functional assessment
Exercise adherence	Extent to which participants perform the prescribed exercise programme.	Exercise monitoring record

5.5.4 Primary Outcome Measure

Functional ability is suggested to be the main outcome since the study is to assess functional performance improvement after the home-based exercise intervention. Functional ability will be assessed by the disability questionnaire outlined in the finalised study protocol.

Pain and discomfort – measured to test if the exercise programme has an effect on the intensity of musculoskeletal discomfort.

Range of motion: Assessed to find out if the exercise intervention increases joint mobility.

Level of functioning: assessed to determine the physical and functional changes experienced by participants.

Functional mobility/work performance: measured to determine alteration in the ability to perform activities with repetitive upper-limb actions.

5.6 Measurement Tools and Materials.

An inclinometer, armchair, measuring tape, tape, stopwatch, pen and paper, and ODI, ROM and QOL are listed as assessment related items in the original IEC proposal. The exact procedures and the final instrument list should be finalized before data collection.

5.6.1 Inclinometer

Test: Range of motion of the joints. Selected joints' range of motion will be measured using an inclinometer. Assessment and positioning will be the same and will be taken using degrees.

5.6.2 Armchair

Result: Participants are positioned in a standard way during assessment. An arm chair will be used to give support and positioning for some assessments.

5.6.3 Measuring Tape

To measure appropriate physical dimensions, as needed. A measuring tape will be used as appropriate for physical measurements as well as for consistency in the assessment procedures.

5.6.4 Stopwatch

Application: Timed Testing procedures and Exercise activity. A stopwatch will be utilized when time measurement is needed during assessment and/or intervention.

5.6.5 Pen and Paper

Target: To gather information about the participants and the results of the assessments. Information on participants, assessment results, adherence of exercises and observations will be recorded using these materials.

5.6.6 Assessment Questionnaires

Purpose: Disability assessment and quality of life. The initial idea mentions the Oswestry Disability Index (ODI), quality of life and range of motion. An exact quality of life instrument and pain assessment instrument need to be identified prior to executing the final protocol.

Oswestry Disability Index is a methodologically clarified instrument that mainly assesses disability due to low-back. The use of this study as the main outcome in the upper-limb musculoskeletal disorders is limited and needs to be confirmed by the research supervisor. Any replacement using a measure specific to the upper limb should only be made at the appropriate revision and approval of the protocol.

5.7 Procedure

The researcher must obtain ethical approval and institutional permission before conducting the research.

Prior to starting the research, the Institutional Ethics Committee of ABSMARI will be consulted. Skill Development Institute, Bhubaneswar will also be approached and eligibility of the participants will be sought with their permission. The study will not start until the necessary approvals have been received.

To recruit and screen participants.

Tailors who are engaged in machine-based tailoring will be approached in the study setting. The purpose, objectives, duration and procedures will be discussed. The inclusion and exclusion criteria will be used to screen possible participants.

5.7.3 Informed Consent

Consent will be obtained in a written informed consent prior to the assessment for research. Participants will be told what the study is about, how long it will last, how it will be conducted, the benefits and risks, and that the project is voluntary and they can withdraw at any time without penalty and their personal information will be kept confidential.

5.7.4 Baseline Assessment

Baseline assessment will consist of the demographic information, occupational information, relevant medical and musculoskeletal history, pain and discomfort, functional ability, range of motion of selected affected joints and quality of life as measured by selected standardized instrument. Results will be documented on the Case Record Form.

5.7.5 Group Allocation

The participants will then be randomly assigned to the experimental or control group after the baseline assessment. If the proposal makes reference to an RCT, then allocation should be done by a documented randomization procedure. The randomization sequence and the allocation ratio and concealment of allocation should be decided prior to enrolment.

5.7.6 Intervention Delivery

The experimental group will be given details of the structured home-based exercise programme. The exercises will be modeled and explained by the investigator. There will be an illustrated booklet or video guide available for the participants. The programme will feature all the exercises from the aerobic, resistance and flexibility group for the neck, shoulder, scapular muscles, upper limbs, wrists and forearms.

5.7.7 Monitoring during the intervention

The participants will be followed for the 8-week period of the intervention. Performance and compliance with exercise will be recorded. Participants will be asked to report any unusual pain, unusual fatigue, muscle cramps or muscle discomfort. Adverse events will be reported and properly assessed.

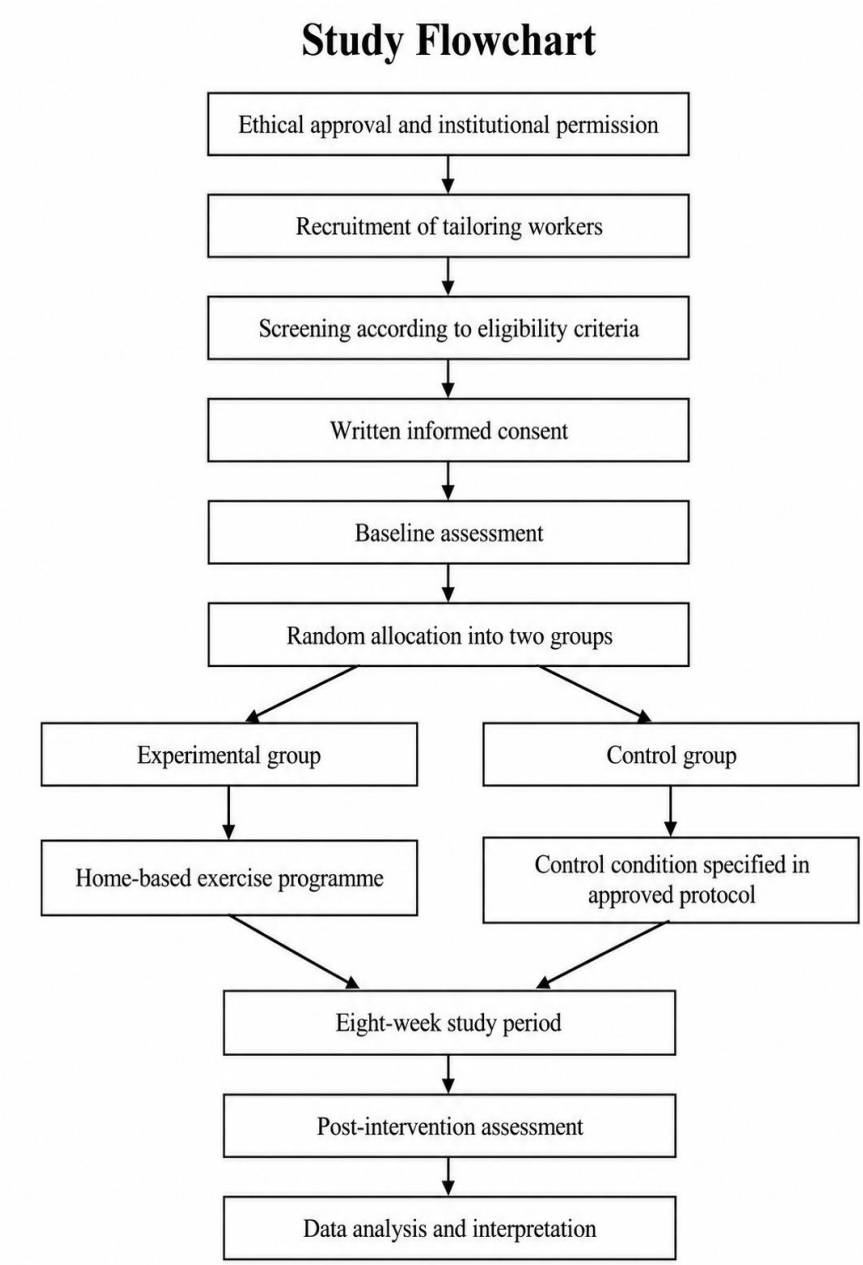
5.7.8 Post-Intervention Assessment

Participants will be assessed at the end of the 8-week intervention on the same measures as at baseline. Pre and Post intervention findings will be compared to determine change.

5.7.9 Data Recording and Management

The case record form will be used to document all information. Each participant will receive an ID number. Data will be entered into Microsoft Excel and then used in the appropriate data analysis software. Analytical dataset to not contain personal identifiers.

5.7.10 Study Flow



5.8 Training / Experimental Protocol

5.8.1 Description of the Intervention

The experimental intervention will include an 8-week structured home-based exercise program to treat work-related upper limb musculoskeletal complaints amongst tailors. Aerobic, resistance and flexibility training will be integrated in this programme focusing on the cervical region, shoulder complex, scapular stabilizers, upper limbs, wrists and forearms. Participants will be provided with an exercise booklet or instructions through video to aid in performing at home.

5.8.2 Duration and Phases of the Intervention

Phase	Duration	Main emphasis
Introductory phase	Weeks 1–4	Familiarization, correct technique, movement coordination and developing exercise tolerance.
Improvement phase	Weeks 5–8	Improving muscular performance, mobility, exercise capacity and adherence.

Exercise progression will be based on individual tolerance and the approved intervention protocol. The original proposal describes the two phases but does not fully specify frequency, session duration or progression criteria; these details must be finalized before implementation.

5.8.3 Aerobic Exercise Programme

Arm circles: Participants will stand or sit comfortably, and perform circular movements with both arms, forwards and backwards. Circles will start with small movements then gradually increase. Prescription: 30 seconds forward and 30 seconds backwards.

Shadow boxing: Participants will stand or sit comfortably with controlled alternating forward or up punching.

Arm marching: Participants will be seated or standing comfortably, and perform arm movements alternatively in a rhythmic marching pattern.

Overhead arm raises: Participants will move the arms forward and out with ease and not with an over-elevated shoulder or painful arm movements.

Towel pull/band rows: hold towel or band with both hands and pull arms apart or row back to the end of the band while emphasizing scapular retraction.

Wrist and finger aerobics: Participants will open and close fists, circles wrists both ways, gently shake hands and move fingers in a comfortable range.

Cool Down: Slow forward and backward shoulder rolls with deep breaths and controlled exhalation: Participants will do slow forward and back rolls of the shoulders and deep breaths in and out.

5.8.4 Resistance Exercise Programme

Exercise	**Target muscles**	**Prescription**	**Equipment**
Chin tucks (isometric)	Deep neck flexors	Hold 5 seconds × 10 repetitions	None
Shoulder shrugs	Upper trapezius	2 × 15 repetitions	Dumbbells or water bottles
Wall push-ups	Chest and shoulder muscles	2 × 10–15 repetitions	Wall
Scapular squeezes	Rhomboids	Hold 5 seconds × 10 repetitions	None
Shoulder external rotation	Rotator cuff	2 × 10 repetitions	Resistance band
Lateral arm raises	Deltoid muscles	2 × 10–15 repetitions	Bottles or resistance band

5.8.5 Flexibility Exercise Programme

Exercise	**Target area**	**Prescription**

Neck side bends	Cervical lateral flexors	Hold 30 seconds × 2 per side
Neck flexion and extension	Anterior and posterior neck muscles	Hold 30 seconds
Shoulder rolls	Shoulder girdle and trapezius	10 forward and 10 backward
Wrist flexor/extensor stretch	Forearm muscles	Hold 30 seconds per arm

5.8.6 Exercise Performance Instructions

Chin tucks: Participants will keep their head up and bring their chin back towards the spine but do not tip the head back. The contraction will be held for 5 seconds and repeated 10 times.

Shoulder shrugs with dumbbells or water bottles: Participants will raise both shoulders towards the ears and slowly and controlledly lower the shoulders.

Wall pushups will have participants facing a wall with their hands against it, bend elbows towards wall, and straighten elbows back to original position.

Scapular squeezes – the participants will squeeze the shoulder blades together whilst keeping a good posture. The contraction will last for 5 seconds.

Shoulder external rotation: Participants will hold a resistance band with elbows at the side of the body and rotate the forearms outwards against resistance, keeping the shoulder controlled.

Lateral arm raises: Participants will perform lateral arm raises with light bottles/resistance bands within a comfortable range then slowly lower to starting position.

Neck side bends: Participants will sit or stand upright without turning the head to the side and bend their neck to one side. The stretch will be held 30 seconds on each side for two reps.

Neck movements forward and backwards: Gents will gently bend the head forward and backwards in a comfortable range. Strong movements will be avoided.

Shoulder rolls: Participants will roll their shoulders in a controlled circle, forward and backward, 10 times each.

Wrist flexor and extensor stretch: Participants will extend one arm forward and stretch wrist flexors and extensors with the other hand. Each stretch will be held for 30 seconds per arm.

5.8.7 Progress in the Exercise Programme

Week 1-4 is the Introductory Phase (Phase I).

The first stage will focus on familiarisation with the exercises to be performed, the correct technique, movement coordination, tolerance to simple exercises of the upper limbs and, avoidance of excessive muscle fatigue and the regularisation of exercise participation. Participants will learn to do exercises using proper body alignment and slow movements.

Phase II: Improvement Phase (Weeks 5-8)

During the improvement phase, the focus will be on increasing muscular strength and endurance, as well as improving the mobility of the shoulder and upper limb, improving the performance of the muscles of the scapula, maintaining the flexibility of the cervical and wrist joints and encouraging adherence to exercise. Progression to be determined by person's tolerance and approved intervention protocol.

5.8.8 Exercise Materials

- Resistance bands.
- Use light dumbbells or water bottles.
- Towels.
- Exercise instruction booklet.

- Demonstrations using video, where appropriate.
- These materials are designed to help you to perform exercises at home without the need of specialist physiotherapy equipment.

5.8.9 Precautions for exercising.

- Controlled movements to be used for exercises.
- Participants will not make any abrupt or vigorous movements.
- Exercises will be done within the range of motion that is comfortable for you.
- Too much resistance will be avoided.
- During exercise participants will be taught not to hold their breath.
- Significant pain/discomfort during exercises will be ceased.
- If any participant complains of unusual symptoms, he or she will be told to contact the investigator.
- Appropriate cool down activities will be provided.
- Adverse effects include temporary muscle pain, tiredness and muscle cramps. Unwanted effects will be noted and assessed throughout the intervention.

5.8.10 Exercise Adherence Monitoring

Follow-up of the home-based exercise programme will be conducted during the intervention period. Participants will receive an exercise record to record the date the activity was performed, exercises performed, difficulty of the exercise, exercise any pain or discomfort, and exercises not performed with a reason. Records will be checked by the investigator to determine participation and any problems with the programme.

5.8.11 Control Group Protocol

The control group will be followed as in the final study protocol, approved control condition. The original IEC proposal mentions two study groups but is not clear about the treatment or activity

that is given to the control group. The control intervention, therefore, needs to be validated by the research supervisor prior to recruitment.

5.9 Summary of the Experimental Protocol

Component	Description
Study design	Experimental study / randomized controlled trial (confirm approved design)
Study population	Male and female tailors
Age	20–45 years
Sampling method	Convenience sampling
Study setting	Skill Development Institute, Bhubaneswar
Intervention duration	Eight weeks
Introductory phase	Weeks 1–4
Improvement phase	Weeks 5–8
Experimental intervention	Home-based exercise programme
Exercise components	Aerobic, resistance and flexibility
Outcome domains	Pain, functional ability, ROM and quality of life
Measurement instrument specified	Inclinometer
Additional materials	Armchair, measuring tape, stopwatch, pen and paper
Assessment schedule	Baseline and post-intervention

5.10 Methodological Points Requiring Confirmation Before Final Submission

- Information that is not fully detailed in the original IEC proposal needs to be clarified with the research supervisor and in the case of an IE Committee, the IE Committee prior to the development of the methodology:
- An appropriate effect size, standard deviation, significance level and statistical power are provided in support of the final sample size.
- How many sessions per week and length of each session?
- Clear progression criteria for the introduction & improvement phases.
- Accurate pain and quality of life assessment tools.
- Appropriateness of the Oswestry Disability Index for an upper limb disorders study.
- The exact condition of the control group.
- Randomization method, allocation ratio, and allocation concealment.
- How to track adherence to exercise and deal with missing data.
- Training and review of two experts to verify the exercise programme.

The final thesis methodology must detail the procedures which shall be approved and implemented. If there is a change in protocol, it is important to be documented and approved before it is put in place.

DATA ANALYSIS AND INTERPRETATION

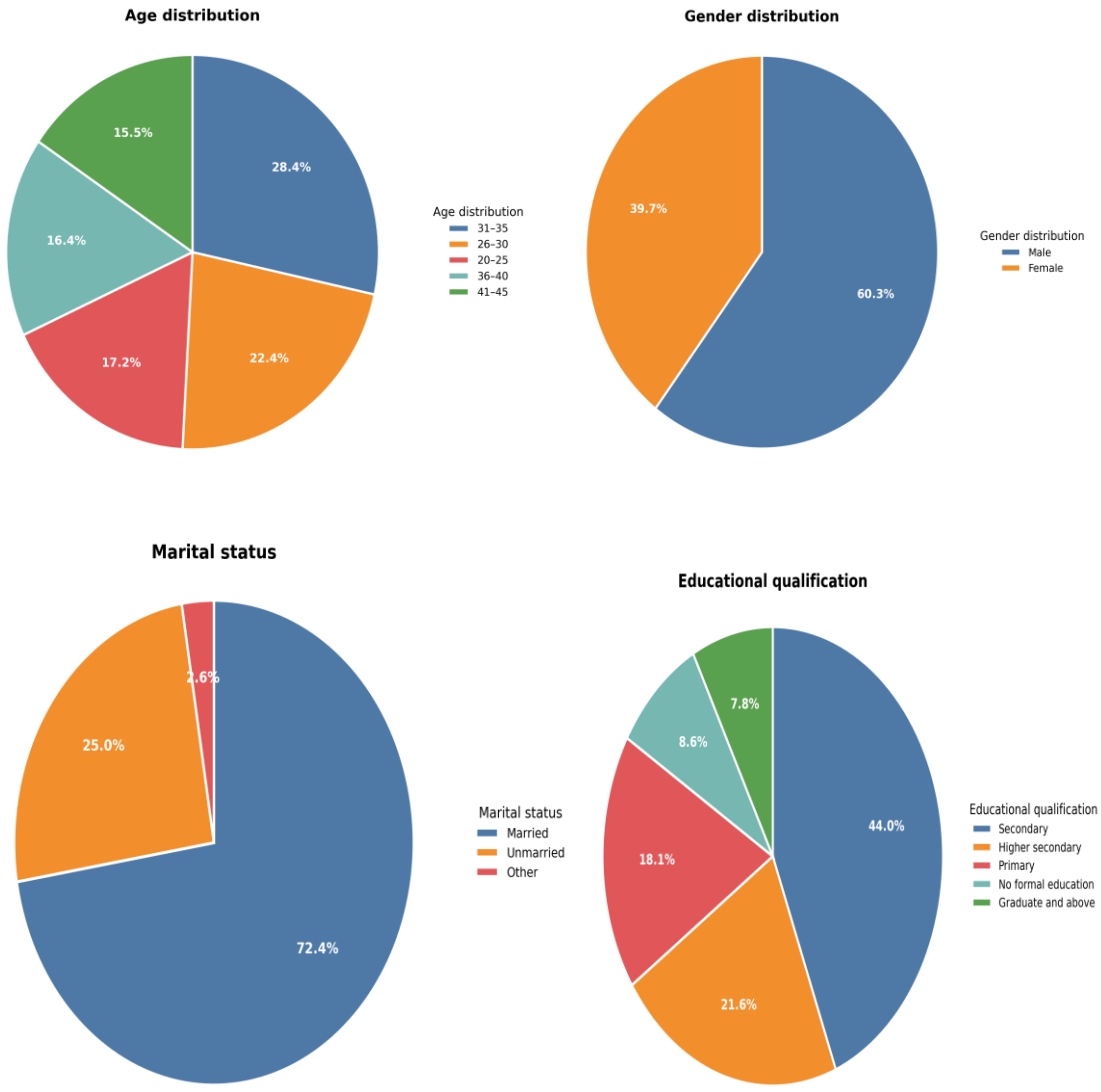
6.1 Demographic and occupational characteristics

Table 6.1. Demographic characteristics of the participants (N=116)

Characteristic	Category	Frequency (n)	Percentage (%)
Age category	31–35	33	28.4
	26–30	26	22.4
	20–25	20	17.2
	36–40	19	16.4
	41–45	18	15.5
Gender	Male	70	60.3
	Female	46	39.7
Marital status	Married	84	72.4
	Unmarried	29	25.0
	Other	3	2.6
Educational qualification	Secondary	51	44.0
	Higher secondary	25	21.6
	Primary	21	18.1
	No formal education	10	8.6
	Graduate and above	9	7.8

Table 6.1 presents the demographic profile of the 116 participants that were included in the study. In terms of age distribution, the highest number of participants were in the 31-35 years age group (28.4%) followed by 26-30 years (22.4%). Participants aged 20-25 years accounted for 20 (17.2%), while 19 (16.4%) were aged 36-40 years. The age group 41-45 years was the lowest age group

with 18 participants (15.5%). This distribution suggests that the study was mostly comprised of people 26-35 years of age.



Regarding sex, males predominated with 70 (60.3%) while females were 46 (39.7%). As far as marital status is concerned, the majority of the participants were married (84; 72.4%) and the next largest group were unmarried (29; 25.0%). Only three (2.6%) participants were in the other marital-status category. In terms of educational qualification, the highest number of respondents was secondary education (51, 44.0%). Fifty-one (43.5%) participants had secondary education and

25 (21.6%) participants had higher secondary education and 21 (18.1%) participants reported primary education. 8.6% (n = 10) participants did not have any formal education and 7.8% (n = 9) participants had graduate level education or higher. Generally, it can be said that a majority of the study population were male, married and secondary-level educated and the age group of 31-35 years was the largest.

6.2 Occupational characteristics

Table 6.2. Occupational characteristics of the simulated participants (N=116)

Occupational variable	Category	Frequency (n)	Percentage (%)
Type of tailoring work	Sewing-machine operator	42	36.2
	Garment stitching	24	20.7
	Independent tailor	23	19.8
	Cutting and stitching	22	19.0
	Other	5	4.3
Professional experience	6–10 years	32	27.6
	11–15 years	30	25.9
	1–5 years	28	24.1
	16–20 years	14	12.1
	Above 20 years	12	10.3
Working hours/day	7–8	43	37.1
	9–10	40	34.5
	Above 10	15	12.9

	4–6	15	12.9
	Below 4	3	2.6
Working days/week	6	72	62.1
	7	24	20.7
	4–5	16	13.8
	1–3	4	3.4
Rest-break frequency	Every 2 hours	38	32.8
	Occasionally	31	26.7
	Every hour	19	16.4
	No regular breaks	17	14.7
	Every 30 minutes	11	9.5
Sewing-machine type	Electric	43	37.1
	Industrial	39	33.6
	Overlock	19	16.4
	Manual	8	6.9
	Other	7	6.0

Table 6.2 indicates the occupational profile of the 116 subjects as regards their nature of tailoring work, professional experience, working hours per day, working days per week, frequency of rest-breaks taken and type of sewing-machine used. In terms of tailoring activities, the category with the greatest number of participants was sewing-machine operators (42 participants, 36.2%). The garment stitching was reported by 24 participants (20.7%), independent tailoring and cutting and stitching by 23 participants (19.8%) and 22 participants (19.0%) respectively. Five participants (4.3%) were engaged in other tailoring related activities. This distribution shows that most of the respondents were directly involved in sewing machine and garment production activities. As to

work experience, the greatest amount of participants had 6-10 years of experience, 32 (27.6%), followed by 11-15 years, 30 (25.9%), and 1-5 years, 28 (24.1%). Fourteen (12.1%) participants had 16-20 years of experience while 12 (10.3%) had more than 20 years of professional experience. In terms of working hours per day, most participants worked 7-8 hours (43; 37.1%) followed by 9-10 (40; 34.5%). Moreover, 15 participants (12.9%) worked 10 or more hours a day and 15 (12.9%) participants worked 4 – 6 hours a day.

When asked about how many days' work a week, 72 (62.1%) participants reported six days; 24 (20.7%) reported working seven days. When it came to frequency of rest breaks, 38 (32.8%) respondents took regularly (every two hours), while 31 (26.7%) took breaks occasionally. The most frequently mentioned equipment was electric sewing machines (43, 37.1%), and then industrial machines (39, 33.6%). In general, the results indicate significant occupational exposure to repetitive sewing tasks, long duration of work and continuous work schedules, which are important contextual characteristics to consider when assessing musculoskeletal symptoms among tailoring workers.

6.3 Baseline group characteristics

Table 6.3. Baseline age and selected work characteristics

Variable	Group/category	Mean or n	SD or %
Age (years)	Experimental	32.22	6.69
	Control	33.12	6.08
Work experience category	6–10 years	32	27.6
	11–15 years	30	25.9
	1–5 years	28	24.1
	16–20 years	14	12.1

	Above 20 years	12	10.3
Daily working hours' category	7–8	43	37.1
	9–10	40	34.5
	Above 10	15	12.9
	4–6	15	12.9
	Below 4	3	2.6

Table 6.3 illustrates the initial age and some selected occupational characteristics of the respondents in the experimental and control groups. The average age of the experimental group was 32.22 +- 6.69 years and the average age of the control group was 33.12 +- 6.08 years. The group means difference was relatively small which indicates that there was a similar age distribution across the groups prior to the intervention. Standard deviations show that there are differences in age for both of these groups, allowing for the inclusion of tailoring workers from varying age groups. When it came to professional experience, the majority of the participants were in the 6-10 years experience category with 32 participants (27.6%). This was followed by participants with 11-15 years of experience (30; 25.9%) and 1-5 years of experience (28; 24.1%). Fourteen individuals (12.1%) had 16-20 years of experience and 12 participants (10.3%) had over 20 years of experience. This distribution suggests that both relatively new and long-term occupationally exposed workers were included in the study population.

Regarding the time spent at work each day, 43 (37.1%) participants said they were working for 7-8 hours, the most common answer. Thirty-four and a half percent (40) of the participants worked 9 to 10 hours a day. Additionally, 15 participants (12.9%) worked for more than 10 hours and 15 worked for 4-6 hours. Three (2.6%) reported working less than 4 hours a day. In sum, the results indicate that tailors were mostly working-age and had extensive experience of tailoring, and were exposed to tailoring on a very regular basis. Monitoring changes in pain, upper-limb functional

limitations, and work-related impact after the intervention in the context of these baseline characteristics can be helpful.

6.4 Objective 1: Pain and musculoskeletal symptoms

Table 6.4. Baseline prevalence of frequent pain and symptom endorsements

Symptom domain	Agree/strongly agree n	%	Neutral/disagree n	%
Neck	68	58.6	48	41.4
Shoulders	59	50.9	57	49.1
Arms/elbows	58	50.0	58	50.0
Wrists/hands	58	50.0	58	50.0
Prolonged-work pain	59	50.9	57	49.1
Post-work stiffness	61	52.6	55	47.4

Table 6.4 shows the base rates of the common musculoskeletal symptoms reported by the tailoring workers in six anatomical symptom domains and six occupational symptom domains. The highest prevalence reported was for pain related to neck with 68 participants (58.6%) answering agree/strongly agree. The other 48 participants (41.4%) answered the questions as neutral, disagree or strongly disagree. This result indicates that neck pain was a major musculoskeletal complaint of the study participants prior to the intervention. 59 of the participants (50.9%) indicated symptoms related to the shoulder and 57 participants (49.1%) indicated neutral or disagree answers. Likewise, 59 (50.9%) participants reported having prolonged-work related pain, meaning that about half of the subjects admitted that they experienced more musculoskeletal pain when they worked for longer periods. Sixty-one of the participants (52.6%) reported having muscular stiffness after work, while 55 (47.4%) reported neutral or disagree. This finding suggests that the

muscular discomfort that arises after the daily work of tailoring was also an issue that was widely reported.

As far as the arm and elbow symptoms are concerned, there were 58 participants (50.0%) who agreed or strongly agreed and 58 participants (50.0%) who were either neutral or disagreed. The wrist/hand symptom row also has a prevalence of 50.0% but the frequency for this symptom in the first category is not provided in the table. The overall findings suggest that neck pain, post-work stiffness and prolonged-work-related discomfort were musculoskeletal issues that were important to participants. The importance of assessing pain-related symptoms and upper limb function when considering tailoring of workers is highlighted by these baseline observations. The results can also be used to investigate if the home-based exercise programme is correlated with the improvements in musculoskeletal health post the intervention period.

Table 6.5. Response distribution for Pain and Musculoskeletal Symptoms: pain and musculoskeletal symptoms (baseline)

Full questionnaire statement	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I experience pain in my neck during or after tailoring activities.	0 (0.0%)	7 (6.0%)	41 (35.3%)	53 (45.7%)	15 (12.9%)
I experience pain in my shoulders during or after sewing activities.	1 (0.9%)	12 (10.3%)	44 (37.9%)	48 (41.4%)	11 (9.5%)
I experience pain in my arms or elbows while performing tailoring work.	0 (0.0%)	12 (10.3%)	46 (39.7%)	50 (43.1%)	8 (6.9%)
I experience pain or discomfort in my wrists and hands during repetitive sewing activities.	0 (0.0%)	8 (6.9%)	50 (43.1%)	47 (40.5%)	11 (9.5%)
My musculoskeletal pain increases after prolonged working hours.	3 (2.6%)	7 (6.0%)	47 (40.5%)	45 (38.8%)	14 (12.1%)

I experience muscular stiffness or discomfort after completing my daily tailoring activities.	1 (0.9%)	8 (6.9%)	46 (39.7%)	51 (44.0%)	10 (8.6%)

Table 6.5 shows the baseline prevalence of pain and musculoskeletal symptoms by the tailoring workers for six questionnaire statements. As for neck pain, the largest percentage of participants (53, 45.7%) agreed that they had neck pain during and/or after tailoring activities, with 15 (12.9%) strongly agreeing. Forty-one more (35.3%) voted in the middle while seven only disagreed (6.0%). This suggests that neck pain was a significant symptom among the participants. Approximately half of the participants (41.4% agreed; 9.5% strongly agreed) endorsed the presence of shoulder-related pain. Likewise, 50 participants (43.1%) agreed/strongly agreed that they had arm/elbow pain. The present results indicate that discomfort did not affect a single anatomical site of the upper limb but multiple sites.

Concerning wrist and hand pain during repetitive sewing, 47 (40.5%) were neutral, 50 (43.1%) disagreed and 11 (9.5%) strongly agreed. This distribution illustrates the variation in participants' wrist and hand symptom experience. In addition, 47 (40.5%) participants were neutral about the increase in pain caused by working for longer periods of time, and 45 (38.8%) agreed. 51 respondents (44.0%) acknowledged that muscular stiffness occurred after working and 10 (8.6%) strongly agreed. In general, the baseline results show a high level of variation in musculoskeletal symptom experiences, with the neck pain, discomfort and post-work stiffness of the shoulder being significant issues among tailing workers.

Table 6.6. Response distribution for Pain and Musculoskeletal Symptoms: pain and musculoskeletal symptoms (post)

Statement	**Strongly disagree**	**Disagree**	**Neutral**	**Agree**	**Strongly agree**

I experience pain in my neck during or after tailoring activities.	3 (2.6%)	27 (23.3%)	46 (39.7%)	33 (28.4%)	7 (6.0%)
I experience pain in my shoulders during or after sewing activities.	10 (8.6%)	26 (22.4%)	48 (41.4%)	26 (22.4%)	6 (5.2%)
I experience pain in my arms or elbows while performing tailoring work.	9 (7.8%)	26 (22.4%)	48 (41.4%)	29 (25.0%)	4 (3.4%)
I experience pain or discomfort in my wrists and hands during repetitive sewing activities.	8 (6.9%)	23 (19.8%)	49 (42.2%)	31 (26.7%)	5 (4.3%)
My musculoskeletal pain increases after prolonged working hours.	7 (6.0%)	26 (22.4%)	47 (40.5%)	31 (26.7%)	5 (4.3%)
I experience muscular stiffness or discomfort after completing my daily tailoring activities.	6 (5.2%)	30 (25.9%)	46 (39.7%)	29 (25.0%)	5 (4.3%)

Table 6.6 illustrates the distribution of the responses following the intervention related to pain and musculoskeletal symptoms in the 116 participants. As for neck pain, 33 (28.4%) agreed, seven (6.0%) strongly agreed, 27 (23.3%) disagreed and three (2.6%) strongly disagreed. Forty-six participants (39.7%) responded with neutral. This distribution suggests that there was less agreement with the neck-pain statement than in the baseline results. Twenty-six participants (22.4%) agreed and six (5.2%) strongly agreed with shoulder pain. Meanwhile, 26 (22.4%) disagreed and 10 (8.6%) strongly disagreed, while 48 (41.4%) selected neutral. In the case of pain in the arm and elbow, 29 (25.0%) agreed and four (3.4%) strongly agreed, while 35 (30.2%) disagreed or strongly disagreed.

On wrist and hand discomfort, 31 (26.7%) agreed and five (4.3%) strongly agreed. Twenty-six additional participants (22.2%) agreed with the statement, 49 (42.2%) selected neutral, and 31 (26.7%) disagreed or strongly disagreed. 31 (26.7%) agreed and five (4.3%) strongly agreed when asked about pain in relation to working long hours. Lastly, muscular stiffness after work was

recognized by 29 participants (25.0%), five (4.3%) strongly agreed. The results show that overall, there was a move toward decreased agreements and increased disagreements on a number of symptom statements at post-intervention. This trend follows a decrease in musculoskeletal symptoms reported. The response distributions, however, need to be reviewed along with the statistical comparisons by group, to determine the magnitude of the change due to the intervention.

Table 6.7. Pain and Musculoskeletal Symptoms total score by group and assessment time

Group	Assessment	Mean	SD	n
Experimental	Baseline	21.19	2.06	58
Experimental	Post	15.12	2.07	58
Control	Baseline	21.16	1.91	58
Control	Post	20.86	2.05	58

Table 6.7 shows the difference in the total Pain and Musculoskeletal Symptoms scores of participants at baseline and post-intervention for both the experimental and control groups. The mean score for the experimental group was 21.19 +- 2.06 while the mean score for the control group was 21.16 +- 1.91 at baseline. The mean difference was small, suggesting that the musculoskeletal symptoms reported in both groups were similar prior to the intervention. The mean score of the experimental group at the end of the intervention period was 15.12, showing a significant decrease from the pre-test mean score of 21.19, and a mean decrease of 6.07 points. The control group exhibited a slightly lower drop from 21.16 (at baseline) to 20.86 (at post-intervention), and a mean decrease of 0.30 point.

Results of the post intervention show a significant difference between the two groups. A lower mean score for the experimental group means that they experienced a better reported pain and musculoskeletal symptom outcome compared to the other group. The questionnaire items are negatively worded so that a decrease in total score is a decrease in the number of symptoms and

discomfort reported during or after tailoring activities. In general, the results show a significant improvement for the experimental group compared to the control group. This pattern indicates that, among tailoring workers, there was a link between participation in the home-based exercise programme and a decrease in self-reported pain and musculoskeletal symptoms.

Table 6.8. Paired-samples t-test for pain and musculoskeletal symptoms

Group	Baseline mean	Baseline SD	Post mean	Post SD	Mean change	t	df	p-value
Experimental	21.19	2.06	15.12	2.07	-6.07	52.77	57	<0.001
Control	21.16	1.91	20.86	2.05	-0.29	3.08	57	0.003

Table 6.8 shows the statistical significance of the change in Pain and Musculoskeletal Symptoms scores from baseline to post-intervention in the experimental group and control group. The mean score of the experimental group, at baseline, was 21.19 +- 2.06; post intervention mean score was 15.12 +- 2.07, resulting in a mean decrease of 6.07 points. The paired-samples t-test showed that there was a significant difference between the two groups (t = 52.77, df = 57, p less than 0.001). The mean score for the control group showed a slight drop from 21.16 +- 1.91 to 20.86 +- 2.05 or a mean score decrease of 0.29 points. The paired-samples t-test also showed a statistically significant reduction (t = 3.08, df = 57, p = 0.003); however, the reduction was comparatively small.

The comparison of the two groups shows that the degree of improvement was significantly higher in the experimental group. A decrease of 6.07 points in the experimental group and 0.29 points in the control group shows that there was a significant difference in the pattern of change. These results show that musculoskeletal symptom scores significantly decreased in both groups, and that there was a far greater decrease in the scores in the group undergoing the home-based exercise programme. The findings highlight the need to consider the impact of the intervention using a comparison between groups rather than just within group statistical significance.

Table 6.9. Paired-samples t-test for upper-limb functional limitations

Group	Baseline mean	Baseline SD	Post mean	Post SD	Mean change	t	df	p-value
Experimental	21.38	1.80	15.33	1.99	-6.05	51.96	57	<0.001
Control	21.52	1.84	21.12	1.97	-0.40	4.04	57	<0.001

Table 6.9 shows the difference in upper limb functional limitation scores before and after the study intervention in both intervention and control groups. The mean score for the experimental group at baseline was 21.38 +- 1.80 while the control group had a slightly higher baseline mean score of 21.52 +- 1.84. The baseline results show the perceived upper-limb functional limitations were similar for both groups prior to the intervention. After the 8-week study period, the mean score of the experimental group had significantly decreased to 15.33 +- 1.99, a mean of 6.05 points less. A paired samples t-test showed a highly statistically significant difference between the pre- and post-test scores ($t = 51.96$, $df = 57$, $p < 0.001$).

By contrast, the control group experienced a relatively small decrease in its mean score, from 21.52 +- 1.84 to 21.12 +- 1.97, a mean score reduction of 0.40 points. This difference was also statistically significant ($t = 4.04$, $df = 57$, p less than 0.001), however. Results show that the experimental group had significantly more improvement in perceived functional limitations than the control group. A higher score on the questionnaires reflects a more challenging ability to perform upper-limb activities, so a lower score after the intervention reflects a perceived ease in performing the activity. Overall the findings show significant improvement in the upper limb functional limitation scores in the home based exercise programme group. This indicates a positive relationship between the intervention and participants' self-efficacy for repetitive hand movements, overhead reaching and use of tailoring tools and working postures.

Table 6.10. Paired-samples t-test for work-related impact and quality of life

Group	Baseline mean	Baseline SD	Post mean	Post SD	Mean change	t	df	p-value
Experimental	21.48	1.88	15.53	2.19	-5.95	47.96	57	<0.001
Control	21.41	2.00	21.03	2.04	-0.38	3.55	57	<0.001

This change is illustrated in the work-related impact scores and the quality of life scores at baseline and post-intervention for experimental and control subjects in Table 6.10. The experimental group had a mean score of 21.48 (+/-1.88) and the control group had a mean score of 21.41 (+/-2.00) at baseline. That's to show that there was no difference in the level of perceived work-related impact or quality-of-life difficulties in both groups prior to intervention. After the intervention the mean score of the experimental group was 15.53 +- 2.19 which is a mean decrease of 5.95 points. A paired-samples t-test revealed a highly statistically significant difference between the scores before and after the intervention (t = 47.96, df = 57, p less than 0.001).

The control group, on the other hand, showed a relative small decrease, from 21.41 +- 2.00 to 21.03 +- 2.04, with a mean decrease of 0.38 points. However the difference within the groups was significant (t = 3.55, df = 57, p less than 0.001). This significant decrease in the experimental group shows a more positive shift in participants' perceptions of their experiences. The questionnaire statements all relate to negative consequences of musculoskeletal symptoms on daily life, working hours, sleep, social engagement and physical health, with lower scores meaning that musculoskeletal symptoms are not perceived to have a greater effect. The results overall showed that for participants in the experimental group, scores for work-related impact and perceived quality-of-life scores improved significantly when compared to the control group. Results indicate that the home-based exercise programme might have a beneficial effect on decreasing perceived musculoskeletal symptom interference with the occupational and daily activities.

Table 6.11. Baseline upper-limb range-of-motion measurements

Movement	Right mean (°)	Right SD	Left mean (°)	Left SD
Shoulder flexion	145.59	5.82	145.59	5.82
Shoulder extension	47.71	7.04	47.71	7.04
Shoulder abduction	137.23	7.14	137.23	7.14
Shoulder external rotation	71.52	7.09	71.52	7.09
Shoulder internal rotation	66.66	6.64	66.66	6.64
Elbow flexion	134.53	6.24	134.53	6.24
Elbow extension	2.71	3.69	2.71	3.69
Wrist flexion	71.73	6.81	71.73	6.81
Wrist extension	60.67	6.37	60.67	6.37
Wrist radial deviation	18.20	7.11	18.20	7.11
Wrist ulnar deviation	31.43	7.06	31.43	7.06

Table 6.11 shows the pre-intervention programme baseline range of motion of shoulder, elbow and wrist joints of the participants. The results give insight in the initial upper limb mobility of the participants and reference values to assess the changes after 8 weeks of the study. When considering the shoulder mobility, mean shoulder flexion was 145.59 +- 5.82deg and the shoulder extension was 47.71 +- 7.04deg. The mean measurement of the shoulder abduction was 137.23 +- 7.14deg, while the external and internal rotation of the shoulder were 71.52 +- 7.09deg and 66.66 +- 6.64deg respectively. These are baseline values of the shoulder's mobility in various directions.

The mean flexion angle recorded for elbow joint was 134.53 +- 6.24deg and mean extension angle of elbow joint was 2.71 +- 3.69deg. The mean flexion measurement was 71.73 +- 6.81deg and the mean extension measurement was 60.67 +- 6.37deg in regards to wrist mobility. A mean radial deviation of 18.20 ± 7.11° and a mean ulnar deviation of 31.43 ± 7.06° was observed. The baseline measurements are for a number of movements of the upper limbs that are of special relevance to adapting activities and skills such as repetitive sewing or cutting, handling tailoring tools, and sustaining working postures. The measurements serve as a good reference for comparison of joint mobility before and after the intervention. Table 6.11 shows an overall baseline assessment of shoulder, elbow and wrist mobility that can be used to assess for changes in upper-limb ROM at the end of the eight-week study period.

6.5 Objective 4: Upper-Limb Range of Motion

Table 6.12. Post upper-limb range-of-motion measurements

Movement	Right mean (°)	Right SD	Left mean (°)	Left SD
Shoulder flexion	150.38	7.62	150.38	7.62
Shoulder extension	52.43	8.66	52.43	8.66
Shoulder abduction	141.93	8.86	141.93	8.86
Shoulder external rotation	76.21	7.95	76.21	7.95
Shoulder internal rotation	71.66	7.85	71.66	7.85
Elbow flexion	139.67	7.24	139.67	7.24
Elbow extension	7.82	5.78	7.82	5.78

Wrist flexion	76.90	7.95	76.90	7.95
Wrist extension	65.56	8.50	65.56	8.50
Wrist radial deviation	23.18	8.26	23.18	8.26
Wrist ulnar deviation	36.36	8.91	36.36	8.91

Table 6.12 shows post intervention measurement of the range of motion of the upper limb joints (shoulder, elbow and wrist) of the participants in the study and shows an improvement in the value of the mean recorded for each upper limb movement compared with the baseline measurements recorded in Table 6.11. The mean shoulder flexion post-intervention was 150.38deg compared to 145.59deg at baseline, and mean shoulder extension improved from 47.71deg at baseline to 52.43deg post-intervention in regard to shoulder mobility. Likewise, shoulder abduction improved from 137.23deg to 141.93deg while external rotation improved from 71.52deg to 76.21deg. Internal rotation also showed improvement from 66.66deg to 71.66deg, which was good in all directions of shoulder joint movement. Regarding the elbow mobility, the mean value of the elbow flexion was improved from 134.53deg to 139.67deg and the value of elbow extension increased from 2.71deg to 7.82deg which means that the range of elbow movement was improved. The mean wrist flexion was 71.73deg before surgery and 76.90deg after surgery, and wrist extension was 60.67deg before surgery and 65.56deg after surgery. Radial deviation went up by 2.38deg and ulnar deviation went up by 4.93deg. It can be seen that there is a progressive increase in the range of motion recorded for all 11 movements that involved the shoulder, elbow and wrist joints. These movements are important for tailoring workers, since the activities of their professions involve repetitive movements of the upper limbs, hand coordination and position, reaching, sewing, and long duration of working positions. The overall results for Table 6.12 show improved upper-limb

range-of-motion measurements recorded after the study period, which is important for assessing improvements in joint mobility with the home-based exercise programme.

Table 6.13. Pre–post ROM change by movement and study group

Movement	Group	Baseline mean (°)	Post mean (°)	Mean change (°)	Paired t	p-value
Shoulder flexion	Experimental	145.76	154.86	9.09	-34.32	<0.001
Shoulder flexion	Control	145.41	145.91	0.49	-1.84	0.071
Shoulder extension	Experimental	48.30	57.36	9.07	-31.47	<0.001
Shoulder extension	Control	47.13	47.49	0.37	-1.44	0.156
Shoulder abduction	Experimental	138.19	146.65	8.46	-39.39	<0.001
Shoulder abduction	Control	136.28	137.20	0.93	-3.15	0.003
Shoulder external rotation	Experimental	70.89	79.40	8.51	-31.35	<0.001
Shoulder external rotation	Control	72.15	73.01	0.85	-3.26	0.002
Shoulder internal rotation	Experimental	66.08	75.00	8.92	-33.25	<0.001
Shoulder internal rotation	Control	67.24	68.31	1.07	-4.18	<0.001

Shoulder internal rotation						
Elbow flexion	Experimental	134.23	143.16	8.93	-46.49	<0.001
Elbow flexion	Control	134.84	136.19	1.35	-6.38	<0.001
Elbow extension	Experimental	3.15	12.26	9.11	-31.77	<0.001
Elbow extension	Control	2.26	3.37	1.11	-5.09	<0.001
Wrist flexion	Experimental	71.75	81.15	9.39	-39.07	<0.001
Wrist flexion	Control	71.70	72.66	0.96	-4.04	<0.001
Wrist extension	Experimental	61.11	70.35	9.24	-30.78	<0.001
Wrist extension	Control	60.22	60.76	0.53	-1.88	0.065
Wrist radial deviation	Experimental	17.70	26.69	8.99	-36.94	<0.001
Wrist radial deviation	Control	18.70	19.68	0.98	-2.92	0.005
Wrist ulnar deviation	Experimental	32.38	41.49	9.11	-37.02	<0.001
Wrist ulnar deviation	Control	30.49	31.23	0.74	-2.64	0.011

As seen in table 6.13, the upper-limb range of motion pre- to post-intervention changes across 11 movements was greater in the experimental group in all movements, indicating greater joint mobility following the intervention. In the case of shoulder flexion, the experimental group showed a gain from 145.76deg to 154.86deg (9.09deg of improvement) which was statistically significant ($t = -34.32$, p less than 0.001); the control group showed a gain from 146.08deg to 146.56deg (0.49deg of improvement), which was not statistically significant ($p = 0.071$). Likewise, experimental group shoulder extension increased 9.07deg while the control group increased 0.37deg, and was highly statistically significant (p less than 0.001). There were also statistically significant (p less than 0.001) shoulder abduction, external rotation and internal rotation mean changes of 8.46deg, 8.51deg and 8.92deg, respectively in the experimental group. Elbow flexion and extension showed comparable improvement of 8.93deg and 9.11deg respectively which showed highly significant difference between the pre and post measurements. In terms of wrist mobility, the experimental group showed an increase of 9.39deg in flexion, 9.24deg in extension, 8.99deg in radial deviation and 9.11deg in ulnar deviation, all of which were statistically significant ($p<0.001$). The control group showed significantly less improvement in all of the movements assessed, with some not being statistically significant such as shoulder flexion ($p = 0.071$), shoulder extension ($p = 0.156$) and wrist extension ($p = 0.065$). Other movements in the control group also showed statistically significant increases, but were much smaller than those found in the experimental group. Overall results showed significantly greater improvements in shoulder, elbow and wrist mobility among participants in the home-based exercise programme, showing a positive correlation between the home-based exercise programme and the improvement in upper-limb joint mobility pertinent to the activities of tailoring workers.

Table 6.14. Between-group comparison of change in the three questionnaire factors

Outcome	Experimental change mean	SD	Control change mean	SD	Mean difference	Welch t	p-value
Factor 1: Pain/symptoms	-6.07	0.88	-0.29	0.73	-5.78	-38.67	<0.001
Factor 2: Functional limitations	-6.05	0.89	-0.40	0.75	-5.66	-37.12	<0.001
Factor 3: Work impact/QOL	-5.95	0.94	-0.38	0.81	-5.57	-34.04	<0.001

Table 6.14 shows the between group differences in change on the three factors of the questionnaire: Pain and Musculoskeletal Symptoms, Upper-Limb Functional Limitations, and Work-Related Impact and Quality of Life, with significant differences in the reduction of adverse scores on the questionnaires between those in the experimental and control groups. The mean reduction for the experimental group on Factor 1 (Pain and Musculoskeletal Symptoms) was 6.07 +- 0.88 points, while the mean reduction for the control group was 0.29 +- 0.73 points, with a mean difference between groups of -5.78 points. Using the Welch's t-test, there was a highly statistically significant difference between the groups (t = -38.67, p less than 0.001) and a significantly larger reduction in reported musculoskeletal symptoms was seen in the group receiving the intervention. Likewise, the mean change in the experimental group for Factor 2 (Upper-Limb Functional Limitations) was -6.05 +- 0.89 points, while the mean change in the control group was -0.40 +- 0.75 points, resulting in a mean difference of -5.66 points. This difference was significant (t = -37.12, p < 0.001) and is indicative of a significant improvement in perceived upper-limb functional ability in the experimental group. For Factor 3, Work-Related Impact and Quality of Life, the experimental group had a mean score decrease of 5.95 +- 0.94 points, the control group had a mean score

decrease of 0.38 +- 0.81 points, and the difference between the groups was a mean score decrease of -5.57 points. The same Welch's t-test also showed a very high level of statistical significance ($t = -34.04$, $p < 0.001$). This shows that the experimental group had significantly more reductions in all three of the questionnaire factors than the control group. The questionnaire statements are all negatively worded; lower scores indicate fewer reported symptoms, fewer functional limitations and less adverse work-related impact. Table 6.14 shows that, overall, there are statistically significant differences between groups for all three outcome measures, favouring the more positive pattern of change for those in the 8-week home-based exercise group.

Table 6.15. One-way ANOVA of factor-score changes across age categories

Outcome	**F statistic**	**p-value**
Pain and Musculoskeletal Symptoms	0.62	0.647
Upper-Limb Functional Limitations	0.90	0.466
Work-Related Impact and Quality of Life	0.61	0.655

Table 6.15 shows the 1-way analysis of variance (ANOVA) comparing the differences in the changes for the three factors of the questionnaire by age groups of the participants. Analysis examines if age-related differences contributed to differences in Pain and Musculoskeletal Symptoms, Upper-Limb Functional Limitations, Work-Related Impact and Quality of Life outcomes after the study period. For Pain and Musculoskeletal Symptoms, the ANOVA yielded an F-statistic of 0.62 and a p-value of 0.647; there was no significant difference found between the score changes across the different age categories. Likewise, Upper-Limb Functional Limitations had an F-statistic of 0.90 with a p-value of 0.466, indicating that there was no significant difference between the increase in perceived upper-limb functional ability for people in different age groups.

For Work-Related Impact and Quality of Life, analysis showed an F-statistic of 0.61 and a p-value of 0.655 – again, no statistically significant differences between outcome changes across the age categories. All three p-values were greater than the commonly used 0.05, which means that there was insufficient statistical evidence of age-related differences in the outcomes measured. These data indicate that the musculoskeletal symptoms, functional limitations and work-related impact observed in this sample were not statistically different among the age groups analyzed. Table 6.15 shows overall that age did not differ significantly with regard to the differences between the three factors of the questionnaire within the sample analysed, but should be interpreted in view of the distribution and variation of the sample in each age group.

Table 6.16. Weekly exercise adherence in the experimental group

Week	**Participants**	**Mean adherence (%)**	**SD**	**Median (%)**
Week 1	116	87.1	6.8	87.0
Week 2		88.3	6.8	88.0
Week 3		86.1	7.0	87.0
Week 4		85.5	5.7	85.0
Week 5		86.3	7.2	86.0
Week 6		87.7	8.0	88.0
Week 7		87.6	7.5	88.5
Week 8		88.1	7.5	88.5

Table 6.16 shows the adherence of participants in the experimental group to the 8-week home-based exercise programme and their consistency in performing the exercises as suggested during the week. The overall mean adherence to the programme was 87.1% +- 6.8% and median adherence 87.0% in the first week, showing a good level of adherence to the programme from the start. During the second week, the mean adherence was 88.3%, with a median of 88.0% which

constituted the highest average adherence for the eight week period. The adherence levels in the third and fourth weeks were 86.1% and 85.5%, respectively, with the lowest week being the 4th, but still above 85%. The medians of these adherence values were 87.0% and 85.0%, respectively. The mean adherence rose slightly to 86.3% in the 5th month, and 87.7% in the 6th month, with a corresponding median of 86.0% and 88.0% respectively. The mean adherence for the seventh week was 87.6% with a median of 88.5% and for the final week was 88.1% with a median of 88.5%. SDs ranged from 5.7% to 8.0% representing a certain amount of variation in adherence among the participants across the course of the intervention period. However, there was no significant decrease towards the end of the programme, as the weekly mean adherence percentages ranged between 85.5% and 88.3%. Table 6.16 shows that the participation in the intervention and engagement in the prescribed exercises remained relatively stable over the course of the 8-weeks, which indicates that the participants were relatively consistent with respect to adherence to the home based exercise programme. Importantly, this adherence information helps to put into perspective the actual changes that were observed for pain, upper-limb functional limitations and range of motion.

Table 6.17. Internal consistency of the three questionnaire factors (Cronbach's alpha)

Factor	**Items**	**Cronbach's alpha**
Pain and Musculoskeletal Symptoms	6	0.735
Upper-Limb Functional Limitations		0.834
Work-Related Impact and Quality of Life		0.849

Table 6.17 presents the Cronbach's alpha coefficients that indicate the internal consistency and reliability of each of the three factors of the questionnaire, which represent the extent to which the six statements of each construct measure the construct. For the Pain and Musculoskeletal Symptoms factor, internal consistency among the six statements was acceptable, as the Cronbach's alpha coefficient was 0.735, with the statements concerning neck, shoulders, arms, elbow,

wrist/hand discomfort and pain with extended working hours. Likewise, the Upper-Limb Functional Limitations factor had a Cronbach alpha coefficient of 0.834 with good internal consistency amongst the six statements measuring upper limb difficulties for overhead reaching, reaching behind the head or back, repetitive hand/finger movements, using tailoring tools and holding upper limb positions when using a sewing machine. The best internal consistency of the statements was obtained for the Work-Related Impact and Quality of Life factor as measured using the Cronbach's alpha coefficient, with 0.849. All factors of the questionnaire showed alpha above 0.70 when compared to the three coefficients, which shows that the questionnaire has a satisfactory internal consistency. Work-Related Impact and Quality of Life had the highest reliability, whereas Pain and Musculoskeletal Symptoms and Upper-Limb Functional Limitations were the next two factors. These findings show that the items of the questionnaire in each factor have an acceptable to good extent of consistency in measuring the dimensions of each factor. Finally, Table 6.17 confirms the internal consistency of the three factors of the questionnaire and the validity of using these factors to analyse pain and musculoskeletal symptoms, functional limitations and work-related impact among tailing workers.

Table 6.18. Correlation among changes in questionnaire factor scores

Variable 1	**Variable 2**	**Pearson r**	**p-value**
Pain and Musculoskeletal Symptoms	Upper-Limb Functional Limitations	0.928	<0.001
	Work-Related Impact and Quality of Life	0.925	<0.001
Upper-Limb Functional Limitations		0.927	<0.001

Table 6.18 shows that Pearson correlation coefficients between changes in the three factors of the questionnaire (Pain and Musculoskeletal Symptoms, Upper-Limb Functional Limitations, and

Work-Related Impact and Quality of Life) indicate relationships between changes in the various dimensions of the study outcomes. A very strong positive correlation was found between changes in Pain and Musculoskeletal Symptoms and changes in Upper-Limb Functional Limitations (r = 0.928, p less than 0.001), demonstrating that change in perceived upper-limb functional limitations were closely related to changes in reported musculoskeletal symptoms and pain. This indicates that those who had larger scores changes on the symptom measures also had larger scores changes on the functional limitation measures. Likewise, a high positive correlation was found between Pain and Musculoskeletal Symptoms and Work-Related Impact and Quality of Life ($r = 0.925$, $p < 0.001$), showing that changes in the Pain and Musculoskeletal Symptoms correlated closely with changes in the perceived impact of musculoskeletal problems on daily home activities, working hours, sleep and social participation. In addition, the correlation of Upper-Limb Functional Limitations with Work-Related Impact and Quality of Life was also highly positive (r = 0.927; p less than 0.001), suggesting that there was a close relationship between the changes in UFL and WRIQOL. Correlation coefficients were all above 0.90, very strongly positively correlated, all p-values < 0.001, statistically significant. The results indicate high overlap among the three aspects of musculoskeletal health, and that improvements in one dimension are likely to correlate with improvements in the other dimensions. Table 6.18 shows good and statistically significant associations between the changes in pain, and upper-limb functional limitations and work-related impact, suggesting that the questionnaire outcomes are interwoven. These correlations, however, show relationships among factors, and not causal relationships.

Table 6.19. Item-wise pre–post comparison for Pain and Musculoskeletal Symptoms

Full questionnaire statement	Baseline mean	Post mean	Mean change	Paired t	p-value
I experience pain in my neck during or after tailoring activities.	3.66	3.12	-0.53	9.90	<0.001

I experience pain in my shoulders during or after sewing activities.	3.48	2.93	-0.55	9.75	<0.001
I experience pain in my arms or elbows while performing tailoring work.	3.47	2.94	-0.53	9.27	<0.001
I experience pain or discomfort in my wrists and hands during repetitive sewing activities.	3.53	3.02	-0.51	8.96	<0.001
My musculoskeletal pain increases after prolonged working hours.	3.52	3.01	-0.51	10.23	<0.001
I experience muscular stiffness or discomfort after completing my daily tailoring activities.	3.53	2.97	-0.55	9.99	<0.001

Results of Pain and Musculoskeletal Symptoms scores for each item on the questionnaires at baseline and post-intervention indicate that there were reductions in mean scores of the responses on all symptom areas assessed (Table 6.19). For tailoring activities, the mean score for neck pain decreased from 3.66 at baseline to 3.12 after the intervention, with a statistically significant mean decrease of 0.53 points (t = 9.90, p less than 0.001). In a similar way, there was a decrease in shoulder pain from 3.48 to 2.93, with a mean decrease of -0.55 points (t = 9.75, p less than 0.001), which is a good decrease in shoulder discomfort. There was also a significant decrease in arm or elbow pain during tailoring activities (3.47 to 2.94, a reduction of 0.53 points, t = 9.27, p less than 0.001). Moreover, wrist/hand discomfort while doing repetitive sewing activities went down from 3.53 to 3.02 (mean = 0.51 points, t = 8.96 and $p<0.001$). The pain scores for prolonged work time (3.52 to 3.01) and muscular stiffness after work (3.53 to 2.97) both significantly decreased (p less than 0.001). The biggest mean changes reported were for shoulder pain and muscular stiffness after working of about 0.55 points. The results show beneficial changes in several anatomical areas, such as the neck, shoulders, arms, elbows, wrists and hands, which are relevant for the repetitive work tasks of tailoring workers. Table 6.19 shows statistically significant decreases on all 6 items of the pain and musculoskeletal symptom questionnaire, indicating that participants

self-reported less musculoskeletal symptoms after the intervention period, as evidenced by lower scores on the negatively worded symptom statements.

Table 6.20. Item-wise pre–post comparison for Upper-Limb Functional Limitations

Full questionnaire statement	Baseline mean	Post mean	Mean change	Paired t	p-value
I experience difficulty raising my arms above shoulder level.	3.64	3.08	-0.56	11.01	<0.001
I experience difficulty reaching behind my head or back.	3.60	3.10	-0.50	9.25	<0.001
I experience difficulty performing repetitive hand and finger movements.	3.60	3.03	-0.58	11.07	<0.001
I experience difficulty holding and handling tailoring tools for prolonged periods.	3.47	2.96	-0.52	9.11	<0.001
I experience difficulty maintaining my upper-limb position while operating a sewing machine.	3.49	2.96	-0.53	10.17	<0.001
My upper-limb discomfort limits my ability to complete tailoring activities efficiently.	3.64	3.10	-0.53	9.90	<0.001

Table 6.20 shows the results of the item-by-item comparison of Upper-Limb Functional Limitations at the pre and post measurement points for 6 questionnaire statements, which shows the favourable changes in participants' perceived ability to perform upper limb tasks related to tailoring work. In terms of difficulty raising arms above the shoulder level, the mean score at baseline was 3.64 and at post-intervention was 3.08, indicating a change of -0.56 points, which was statistically significant ($t = 11.01$, $p < 0.001$). Difficulty reaching behind the head or back also showed a decline from 3.60 to 3.10 (mean change -0.50 points, $t = 9.25$, p less than 0.001). The reported mean difference in difficulty performing repetitive hand/finger movements was 0.58 points (baseline mean = 3.61 points; post-measurement = 3.03 points) and was statistically

significant (t = 11.07, p less than 0.001). In addition, the number of points indicating difficulty holding and using tailoring tools for longer periods of time reduced from 3.47 to 2.96 (t = 9.11, p less than 0.001). There was also improvement with the difficulty of maintaining upper-limb positions and the mean score dropped from 3.49 to 2.96 representing a reported mean change of -0.53 points (t = 10.17, p less than 0.001). Lastly, the deficiencies in completing tailoring activities efficiently also decreased, from 3.64 to 3.10, with a mean decrease of 0.53 points (t = 9.90, p less than 0.001). These results show improvements in various aspects of upper limb function, including ability to reach overhead and behind the head or back, repetitive hand movements, using occupational tools and maintaining the proper upper limb position when performing sewing tasks. Overall, Table 6.20 shows statistically significant decreases for all six functional limitation statements, indicating that there was a statistically significant change in the perceived upper-limb functional abilities of the participants after the intervention period.

Table 6.21. Item-wise pre–post comparison for Work-Related Impact and Quality of Life

Full questionnaire statement	Baseline mean	Post mean	Mean change	Paired t	p-value
My musculoskeletal symptoms interfere with my daily household activities.	3.68	3.14	-0.54	10.07	<0.001
My physical discomfort affects my ability to complete my regular working hours.	3.58	3.03	-0.55	10.24	<0.001
My musculoskeletal symptoms affect my sleep and rest.	3.47	2.96	-0.52	9.57	<0.001
My physical discomfort reduces my ability to participate in social and recreational activities.	3.59	3.04	-0.55	9.53	<0.001
I feel physically exhausted after performing my regular tailoring activities.	3.47	3.00	-0.47	8.78	<0.001

My musculoskeletal problems reduce my overall satisfaction with my physical health.	3.65	3.12	-0.53	10.00	<0.001

Table 6.21 shows the mean response scores on the six questionnaire statements before and after the intervention period, indicating that the mean scores for each dimension of assessment decreased after the intervention. In terms of the interference of musculoskeletal symptoms with daily activities in the home, the mean score was 3.68 at baseline and 3.14 post-intervention with a mean decrease of 0.54 points, which was statistically significant (t = 10.07, p less than 0.001). Likewise, a mean difference of -0.55 points (t = 10.24, p less than 0.001) was observed for the effect of physical discomfort on the ability to complete regular working hours, reflecting a positive change in participants' perceptions. The mean score for musculoskeletal symptoms affecting sleep and rest also improved from 3.47 to 2.96 (a change of -0.52 points, t = 9.57, p less than 0.001). Similarly, the physical discomfort's effect on social and recreational activities involvement was reduced from 3.59 to 3.04, and the difference was statistically significant (t = 9.53, $p < 0.001$). Physical exhaustion after performing the daily tailoring activities showed a mean difference of -0.47 points (t = 8.78, p less than 0.001) ranging from 3.47 to 3.00. Lastly, the mean reduction from 3.65 to 3.12 in dissatisfaction with overall physical health associated with musculoskeletal problems was significant (t = 10.00, p less than 0.001). The greatest changes were reported for the impact of physical discomfort on normal work hours (change of about 0.55 points) and social and recreational activities (change of about 0.55 points). These results show positive alterations in the perception of musculoskeletal symptoms interfering with the housework, work, sleep, social life and physical health. Table 6.21 shows statistically significant improvements in all six work-related impact and quality-of-life statements, which indicates that the participants reported less interference from musculoskeletal symptoms in their daily and work activities after the intervention time period.

Table 6.22. One-way ANOVA of factor changes by working hours per day

Outcome	F statistic	p-value	Categories analysed
Pain and Musculoskeletal Symptoms	1.09	0.365	5
Upper-Limb Functional Limitations	0.91	0.463	5
Work-Related Impact and Quality of Life	1.08	0.369	5

The one-way analysis of variance (ANOVA) results that investigated the differences in changes between the three questionnaire factors based on the five categories of working hours are presented in Table 6.22 to determine if there were differences in extent of change in musculoskeletal symptoms, functional limitations and work-related impact associated with daily occupational exposures. For Pain and Musculoskeletal Symptoms, no statistically significant difference was found among daily working-hour groups (F-Statistic of 1.09 and p-value of 0.365). This indicates that the number of hours that participants worked each day was not a significant difference in the observed changes in pain and symptom scores. Likewise, the F-statistic for Upper-Limb Functional Limitations was 0.91 and the p-value was 0.463, which means that any differences in perceived upper-limb functional ability were not statistically significant across the five categories of working hours. With respect to Work-Related Impact and Quality of life, the F-statistic was 1.08 with a corresponding p-value of 0.369, and there were no statistically significant differences between outcome changes based on the number of hours worked per day. All three p-values were above the more commonly used threshold for statistical significance of .05, so there was not enough evidence for differences in changes across the three working-hour categories. These findings indicate that the changes in pain, functional limitations and work-related impact

experienced by the participants with the different daily work schedules were relatively similar. Table 6.22 indicates overall that there were no statistically significant differences between the five working-hour categories with regards to the three questionnaire factors, although this may be due to the number of participants in each of the categories and to the variation of individual responses.

Table 6.23. One-way ANOVA of factor changes by work experience

Outcome	F statistic	p-value	Categories analysed
Pain and Musculoskeletal Symptoms	0.69	0.599	5
Upper-Limb Functional Limitations	0.71	0.584	5
Work-Related Impact and Quality of Life	1.02	0.399	5

Table 6.23 emphasizes the results of the one-way analysis of variance (ANOVA) comparing changes across the three factors of the questionnaire as a function of professional experience in tailoring activities, with five categories of professional experience to explore whether duration of occupational experience was related to the changes of pain, functional limitations and work-related impact. The analysis yielded an F-statistic of 0.69 and a p-value of 0.599 for Pain and Musculoskeletal Symptoms, suggesting that there were no significant differences in the change in musculoskeletal symptom scores across the five professional-experience categories. In the same manner, Upper-Limb Functional Limitations had a F value of 0.71 with a p-value of 0.584, and indicated that while there was some increased level of perceived upper-limb functional ability, it was not significantly different based on participants' duration of occupational experience. For Work-Related Impact and Quality of Life, the F-statistic was 1.02 and the p-value was 0.399, also showing no significant differences between change in scores in the five categories of experience.

All three p-values tested were greater than 0.05, thus there was no strong statistical evidence of differences in changes in outcome based on professional experience. It is concluded from these results that the changes in musculoskeletal symptoms, functional limitation and work-related impact measured in tailoring workers were generally similar between workers with different occupational experience. The findings of the analysis, however, do not imply that professional experience is without effect on musculoskeletal health; rather, the results suggest that the differences between the experience categories in the present sample were not seen to be statistically significant in the measured changes in health. Table 6.23 overall shows that within the analysed sample there was no significant difference between changes in the three questionnaire factors based on differences in professional experience.

Table 6.24. One-way ANOVA of factor changes by rest-break frequency

Outcome	**F statistic**	**p-value**	**Categories analysed**
Pain and Musculoskeletal Symptoms	0.17	0.953	5
Upper-Limb Functional Limitations	0.36	0.835	5
Work-Related Impact and Quality of Life	0.38	0.821	5

The one-way analysis of variance (ANOVA) results comparing differences in changes across the three questionnaire factors based on the participants' rest-break frequency during tailoring activities is presented in Table 6.24 and included five categories of rest-break frequency to assess whether the level of improvement in musculoskeletal symptoms, upper-limb functional limitations and work-related impact differed based on participants' rest pattern. The analysis for Pain and Musculoskeletal Symptoms yielded an F-statistic of 0.17 and p-value of 0.953, and revealed no

statistically significant difference in the change in symptom scores between the various rest-break frequencies. In a similar fashion, the F-statistic for Upper-Limb Functional Limitations was 0.36 with p=0.835 indicating that there was no statistically significant difference between the five rest-break categories in terms of upper-limb functional ability perceived change. In the analysis of Work-Related Impact and Quality of Life there was also no statistically significant differences in outcome changes by rest-break frequency, as the F-statistic was 0.38 and p-value was 0.821. All three p-values were above the typical level of significance (p=0.05), indicating there were no statistically significant differences in the change between rest-break groups. Together, these results indicate that, overall, participants with differing rest patterns showed similar musculoskeletal symptom and upper-limb functional limitation and work impact changes over the study period. The results should not be interpreted, however, as proof that these rest breaks have no effect on musculoskeletal health – they only consider the differences in the changes of measured outcomes between the 5 categories. Table 6.24 shows, overall, that there was no significant relationship between the changes and the three questionnaire factors in the analysed sample as far as rest-break frequency is concerned.

Table 6.25. Group-by-time interaction (two-time-point change-score equivalent)

Outcome	**F statistic**	**df (numerator, denominator)**	**p-value**
Pain and Musculoskeletal Symptoms	1495.63	1, 114	<0.001
Upper-Limb Functional Limitations	1378.04	1, 114	<0.001
Work-Related Impact and Quality of Life	1158.53	1, 114	<0.001

Table 6.25 displays the group x time interaction effects for the three questionnaire factors: Pain and Musculoskeletal Symptoms, Upper-Limb Functional Limitations and Work-Related Impact and Quality of Life, to determine if differences between experimental and control groups at baseline and post-intervention were significant. In Pain and Musculoskeletal Symptoms, the F-statistic was 1495.63, with degrees of freedom 1 and 114 and a p-value of < 0.001, which was highly statistically significant, suggesting a highly different pattern of change between the two groups. This finding indicates that the decrease in pain and musculoskeletal symptoms was significant between the home-based exercise programme and control group. The F-statistic for Upper-Limb Functional Limitations was also 1378.04 (df = 1, 114, p less than 0.001) and showed that the difference in the change in perceived upper-limb functional ability between the experimental and control groups was statistically significant. For Work-Related Impact and Quality of Life, the analysis showed an F-statistic of 1158.53 (df = 1, 114, p less than 0.001), which indicated a significant difference between the changes in perceived impact of musculoskeletal symptoms on occupational and daily activities. Each of the three interaction effects was found to be statistically significant ($p < .05$). These findings suggest that there was a significant difference in pain, functional limitation and work related impact between two study groups over the time of the intervention. Table 6.25 shows that statistically significant group-by-time interaction effects were seen across all three factors on the questionnaire, with much greater improvements noted among the experimental group participants, which further supports the favourable effects seen with the eight-week home-based exercise programme.

Table 6.26. Comparison of factor-score changes by gender

Outcome	Male mean change	Female mean change	Welch t	p-value

Pain and Musculoskeletal Symptoms	-3.20	-3.15	-0.08	0.934
Upper-Limb Functional Limitations	-3.26	-3.17	-0.15	0.881
Work-Related Impact and Quality of Life	-3.16	-3.17	0.03	0.976

Table 6.26 focuses on the differences in changes in each of the three questionnaire factors between males and females, to determine if the increase or decrease in Pain and Musculoskeletal Symptoms, Upper-Limb Functional Limitations, and Work-Related Impact and Quality of Life were different by gender. In Pain and Musculoskeletal Symptoms, the mean change scores for males and females were -3.20 points and -3.15 points, respectively, with a Welch t-statistic of -0.08 and a p-value of 0.934, suggesting that the difference between the two gender groups was not statistically significant. In the same manner, the mean change for the Upper-Limb Functional Limitations was -3.26 points for the male participants and -3.17 points for the female participants which yielded a Welch t-statistic value of -0.15 and a p-value of 0.881. This result suggests that perceived upper-limb functional limitations had a statistically similar change for both male and female subjects. For Work-Related Impact and Quality of Life, the male participants showed a change of -3.16 points while the female participants showed a change of -3.17 points with Welch t-statistic of 0.03 and p-value of 0.976, which is close to no difference. All three p-values were greater than the traditional level of significance (0.05) indicating that there were no statistically significant differences in factor-score changes as a function of gender. The results thus indicate that within the sample analysed, males and females showed overall similar changes in musculoskeletal symptoms and functional restrictions and work-related consequences. Table 6.26 shows that in general, there was no significant difference between genders in the changes for any

of the three factors on the questionnaire, however this should be viewed within the context of the gender distribution of and variability within the study population.

6.15 Range-of-Motion Comparison Between Groups

Table 6.27. Descriptive statistics of factor-score changes by group

Factor	Group	Mean change	SD	n
Pain and Musculoskeletal Symptoms	Experimental	-6.07	0.88	58
	Control	-0.29	0.73	58
Upper-Limb Functional Limitations	Experimental	-6.05	0.89	58
	Control	-0.40	0.75	58
Work-Related Impact and Quality of Life	Experimental	-5.95	0.94	58
	Control	-0.38	0.81	58

The descriptive comparison of the mean change across the three questionnaire factors for the experimental and control groups is illustrated in Table 6.27 which gives an overview of the size and range of change that occurred after the eight week study period. The mean change in Pain and Musculoskeletal Symptoms scores for the experimental group in the study showed a reduction of 6.07 +- 0.88 points (n = 58), while the control group showed a much smaller reduction of 0.29 +- 0.73 points (n = 58), suggesting a greater improvement in musculoskeletal symptoms in the experimental group. Likewise, in the case of Upper-Limb Functional Limitations, the mean score of the experimental group showed a decrease of 6.05 +- 0.89 points while the control group showed a decrease of 0.40 +- 0.75 points indicating significantly more improvement in upper-limb functional ability in the home-based exercise group. In Work-Related Impact and Quality of Life the experimental group had a mean decrease of 5.95 +- 0.94 points compared to a mean decrease of 0.38 +- 0.81 points in the control group, which represents a significant reduction of

musculoskeletal symptoms impacting daily and occupational activities among participants in the experimental group. The standard deviations suggest the variability of the individual score changes within each group, and the consistent differences in mean changes suggest a similar pattern for all three questionnaire factors. The changes of means on the negatively worded questionnaire statements are a reduction in adverse symptoms, functional difficulties and work-related impact. Table 6.27 shows that the total score along with the scores of the three factors of the questionnaire changed favourably after the home-based exercise programme with a general trend of more improvement in the experimental group regardless of their condition.

Table 6.28. Between-group comparison of selected ROM changes

Movement	**Experimental mean change**	**Control mean change**	**Mean difference**	**Welch t**	**p-value**
Shoulder flexion	9.10	0.50	8.60	22.77	<0.001
Shoulder abduction	8.46	0.93	7.53	20.70	<0.001
Shoulder external rotation	8.51	0.86	7.65	20.33	<0.001
Elbow flexion	8.94	1.35	7.59	26.53	<0.001
Wrist extension	9.24	0.53	8.71	20.97	<0.001

Table 6.28 illustrates the difference between the groups in the amount of change in range of motion in 5 selected upper limb movements, and shows between group differences in the amount of improvement in the range of motion in these movements between the home-based exercise programme and the control group. As to shoulder flexion, the mean difference was 9.10deg for the

experimental group, and 0.50deg for the control group, for a mean difference of 8.60deg. The t-test showed that the difference between the groups was highly statistically significant (t = 22.77, p less than 0.001), showing that there was significantly more improvement in the shoulder flexion of the experimental group. Shoulder abduction also showed a mean difference of 7.53deg (t = 20.70, p less than 0.001), with the experimental group showing a mean improvement of 8.46deg and the control group showing a mean improvement of 0.93deg. The experimental group had a mean increase of 8.51deg while the control group had a mean increase of 0.86deg when measured externally at the shoulder (mean difference was 7.65deg, t = 20.33, p less than 0.001). For elbow flexion, the mean improvement of experimental group was 8.94deg, and that of control group was 1.35deg, which resulted in a mean difference of 7.59deg (t = 26.53, p less than 0.001). Wrist extension also showed a significantly larger change in the experimental group (mean difference = 9.24deg) compared to the control group (mean difference = 0.53deg), a difference of 8.71deg (t = 20.97, p less than 0.001). Statistically significant differences between groups were shown with all five comparisons with a p value < 0.001 showing significant differences in the amount of improvement in selected shoulder, elbow and wrist movements between the experimental and control groups. These movements have special significance for adapting activities that involve repetition of reaching, position of upper limbs, sewing and coordinated movements of joints. Table 6.28 shows that overall, all five measurements of ROM showed significantly greater improvements in participants who received the home-based exercise programme, indicating a positive correlation between the programme and upper-limb joint mobility.

Table 6.29. Distribution of upper-limb range-of-motion outcomes at baseline and post-intervention

Joint	Movement	Side	Baseline (°), mean ± SD	Post-intervention (°), mean ± SD	Mean difference (°)
Shoulder	Flexion	Right	145.59 ± 5.82	150.38 ± 7.62	4.79
Shoulder	Flexion	Left	145.59 ± 5.82	150.38 ± 7.62	4.79
Shoulder	Extension	Right	47.71 ± 7.04	52.43 ± 8.66	4.72
Shoulder	Extension	Left	47.71 ± 7.04	52.43 ± 8.66	4.72
Shoulder	Abduction	Right	137.23 ± 7.14	141.93 ± 8.86	4.69
Shoulder	Abduction	Left	137.23 ± 7.14	141.93 ± 8.86	4.69
Shoulder	External rotation	Right	71.52 ± 7.09	76.21 ± 7.95	4.68
Shoulder	External rotation	Left	71.52 ± 7.09	76.21 ± 7.95	4.68
Shoulder	Internal rotation	Right	66.66 ± 6.64	71.66 ± 7.85	5.00
Shoulder	Internal rotation	Left	66.66 ± 6.64	71.66 ± 7.85	5.00
Elbow	Flexion	Right	134.53 ± 6.24	139.67 ± 7.24	5.14
Elbow	Flexion	Left	134.53 ± 6.24	139.67 ± 7.24	5.14
Elbow	Extension	Right	2.71 ± 3.69	7.82 ± 5.78	5.11

Elbow	Extension	Left	2.71 ± 3.69	7.82 ± 5.78	5.11
Wrist	Flexion	Right	71.73 ± 6.81	76.90 ± 7.95	5.18
Wrist	Flexion	Left	71.73 ± 6.81	76.90 ± 7.95	5.18
Wrist	Extension	Right	60.67 ± 6.37	65.56 ± 8.50	4.89
Wrist	Extension	Left	60.67 ± 6.37	65.56 ± 8.50	4.89
Wrist	Radial deviation	Right	18.20 ± 7.11	23.18 ± 8.26	4.98
Wrist	Radial deviation	Left	18.20 ± 7.11	23.18 ± 8.26	4.98
Wrist	Ulnar deviation	Right	31.43 ± 7.06	36.36 ± 8.91	4.93
Wrist	Ulnar deviation	Left	31.43 ± 7.06	36.36 ± 8.91	4.93

Table 6.29 shows the comparisons between the range-of-motion measurements at baseline and at the end of the intervention, showing that the range of motion recorded at the end of the intervention is increased in all the 11 movements assessed at all three joints (the shoulder, elbow and wrist). In terms of shoulder mobility, there was an improvement of 4.79 +- 5.82deg from pre to post intervention in the flexion range of motion. In a similar manner, the shoulder extension went up from 47.71 +- 7.04deg to 52.43 +- 8.66deg, with a 4.72deg change. The angle of shoulder abduction increased by 4.69deg, whereas the angle of external rotation increased by 4.69deg, from 71.52 +- 7.09deg to 76.21 +- 7.95deg. There was a 5.00deg increase in the internal rotation from 66.66 +- 6.64deg to 71.66 +- 7.85deg. In terms of elbow mobility, elbow flexion improved from 134.53 +- 6.24deg to 139.67 +- 7.24deg (difference of 5.14deg) and elbow extension improved from 2.71 +- 3.69deg to 7.82 +- 5.78deg (difference of 5.11deg). Regarding wrist movements, flexion improved from 71.73 +- 6.81deg to 76.90 +- 7.95deg, while extension increased from 60.67 +- 6.37deg to 65.56 +- 8.50deg. Radial deviation and ulnar deviation also showed an improvement

of 4.98deg and 4.93deg respectively. These changes suggest there were positive changes in the range of the mean recorded movement for all upper limb movements including shoulder rotation, elbow mobility and wrist movement. These enhancements are applicable to the modulation of activities that involve repetitive movement of the joints, prolonged positioning of the upper limbs, reaching, and coordinated hand function. Table 6.29 shows descriptive data on changes in upper-limb mobility, with an increase in mean ROM scores for all 11 movements that were assessed following the study period.

Table 6.30. Comparison of range-of-motion outcomes between experimental and control groups

Joint	Movement	Study group	Baseline (°), mean ± SD	Post-intervention (°), mean ± SD
Shoulder	Flexion	Experimental	145.76 ± 6.60	154.86 ± 6.85
Shoulder	Flexion	Control	145.41 ± 4.98	145.91 ± 5.42
Shoulder	Extension	Experimental	48.30 ± 7.14	57.36 ± 6.91
Shoulder	Extension	Control	47.13 ± 6.94	47.49 ± 7.34
Shoulder	Abduction	Experimental	138.19 ± 7.16	146.65 ± 7.20
Shoulder	Abduction	Control	136.28 ± 7.05	137.20 ± 7.81
Shoulder	External rotation	Experimental	70.89 ± 6.80	79.40 ± 7.36

Shoulder	External rotation	Control	72.15 ± 7.38	73.01 ± 7.25
Shoulder	Internal rotation	Experimental	66.08 ± 6.62	75.00 ± 6.68
Shoulder	Internal rotation	Control	67.24 ± 6.66	68.31 ± 7.56
Elbow	Flexion	Experimental	134.23 ± 5.75	143.16 ± 5.92
Elbow	Flexion	Control	134.84 ± 6.73	136.19 ± 6.80
Elbow	Extension	Experimental	3.15 ± 3.94	12.26 ± 4.09
Elbow	Extension	Control	2.26 ± 3.40	3.37 ± 3.25
Wrist	Flexion	Experimental	71.75 ± 6.43	81.15 ± 6.34
Wrist	Flexion	Control	71.70 ± 7.24	72.66 ± 7.11
Wrist	Extension	Experimental	61.11 ± 6.82	70.35 ± 7.63
Wrist	Extension	Control	60.22 ± 5.92	60.76 ± 6.38
Wrist	Radial deviation	Experimental	17.70 ± 7.04	26.69 ± 7.34
Wrist	Radial deviation	Control	18.70 ± 7.22	19.68 ± 7.68
Wrist	Ulnar deviation	Experimental	32.38 ± 7.29	41.49 ± 7.49
Wrist	Ulnar deviation	Control	30.49 ± 6.75	31.23 ± 7.12

Table 6.30 indicates the pre- and post-range of motion scores for the experimental and control groups for 11 upper-limb movements, showing a higher increase in the ROM scores recorded for each movement by the experimental group. As for the shoulder flexion, there was an increase from

145.76 +- 6.60deg to 154.86 +- 6.85deg in the experimental group; whereas, the control group exhibited a comparatively small increase from 145.41 +- 4.98deg to 145.91 +- 5.42deg. Likewise, shoulder extension increased by +9.07deg (48.30 to 57.36deg) in the experimental group and by +0.37deg (41.68 to 42.05deg) in the control group. In the shoulder abduction, the experimental group showed their range of motion from 138.19deg to 146.65deg, and the control group showed their range of motion from 136.28deg to 137.20deg. Similarly, external rotation of the shoulder had greater increase in the experimental group (70.89deg to 79.40deg) than the control group (72.15deg to 73.01deg). Likewise, shoulder internal rotation increased by 8.92deg in the experimental group and 1.07deg in the control group. With the elbow mobility, the experimental group showed a gain from 134.23deg to 143.16deg while the control group showed a gain of 134.84deg to 136.19deg. The results also showed that the extension of the elbow had a significantly greater improvement in the experimental group when compared to the control group. Regarding wrist mobility, both flexion and extension along with both radial and ulnar deviation showed greater improvements in the experimental group. These differences show a consistent pattern of increased mobility of the shoulder, elbow and wrist for participants in the home-based exercise programme. Table 6.30 shows that there were more favourable descriptive changes in upper-limb ROM in the experimental group in 11 movements compared with the control group.

Table 6.31. Within-group statistical analysis of pre- and post-intervention range of motion

Joint	Movement	Group	Baseline mean	Post mean	Mean change	Paired t	df	p-value
Shoulder	Flexion	Experimental	145.76	154.86	9.09	-34.32	57	<0.001
Shoulder	Flexion	Control	145.41	145.91	0.49	-1.84	57	0.071
Shoulder	Extension	Experimental	48.30	57.36	9.07	-31.47	57	<0.001
Shoulder	Extension	Control	47.13	47.49	0.37	-1.44	57	0.156
Shoulder	Abduction	Experimental	138.19	146.65	8.46	-39.39	57	<0.001
Shoulder	Abduction	Control	136.28	137.20	0.93	-3.15	57	0.003
Shoulder	External rotation	Experimental	70.89	79.40	8.51	-31.35	57	<0.001

Shoulder	External rotation	Control	72.15	73.01	0.85	-3.26	57	0.002
Shoulder	Internal rotation	Experimental	66.08	75.00	8.92	-33.25	57	<0.001
Shoulder	Internal rotation	Control	67.24	68.31	1.07	-4.18	57	<0.001
Elbow	Flexion	Experimental	134.23	143.16	8.93	-46.49	57	<0.001
Elbow	Flexion	Control	134.84	136.19	1.35	-6.38	57	<0.001
Elbow	Extension	Experimental	3.15	12.26	9.11	-31.77	57	<0.001
Elbow	Extension	Control	2.26	3.37	1.11	-5.09	57	<0.001
Wrist	Flexion	Experimental	71.75	81.15	9.39	-39.07	57	<0.001

Wrist	Flexion	Control	71.70	72.66	0.96	-4.04	57	<0.001
Wrist	Extension	Experimental	61.11	70.35	9.24	-30.78	57	<0.001
Wrist	Extension	Control	60.22	60.76	0.53	-1.88	57	0.065
Wrist	Radial deviation	Experimental	17.70	26.69	8.99	-36.94	57	<0.001
Wrist	Radial deviation	Control	18.70	19.68	0.98	-2.92	57	0.005
Wrist	Ulnar deviation	Experimental	32.38	41.49	9.11	-37.02	57	<0.001
Wrist	Ulnar deviation	Control	30.49	31.23	0.74	-2.64	57	0.011

Table 6.31 shows the statistical significance of the difference between the pre- and post-intervention measurements of upper-limb ROM for the experimental and control groups, respectively, for 11 movements, which were assessed using paired-samples t-tests. Changes in all 11 movements measured were statistically significant for the experimental group. Shoulder flexion increased by 9.09deg (t = -34.32, p less than 0.001), while shoulder extension improved by 9.07deg (t = -31.47, p less than 0.001). The improvements in shoulder abduction and external rotation were

8.46deg and 8.51deg, respectively, which were both statistically significant ($p<0.001$). Likewise, internal rotation of the shoulder increased by 8.92deg ($t = -33.25$, $p < 0.001$). Elbow mobility was evaluated by flexion and extension movements and both flexion (8.93deg) and extension (9.11deg) showed highly statistically significant differences ($p < 0.001$). Wrist mobility also showed significant improvements, with increases of 9.39deg, 9.24deg, 8.99deg and 9.11deg in flexion, extension, radial deviation and ulnar deviation respectively (all four with p values less than 0.001). The relatively smaller changes were noted for the control group for shoulder flexion ($p = 0.071$) and extension ($p = 0.156$). Statistically significant changes were seen in several other movements within the control group; however, the magnitudes of these changes were relatively small compared to the changes in the experimental group. There was no statistically significant difference for wrist extension in the control group ($p = 0.065$). The results indicate statistically significant improvement in all 11 movements examined in the experimental group, while the control group exhibited less and more varied improvements. Table 6.31 shows a positive trend in shoulder, elbow, and wrist mobility in those that participated in the home based exercise programme.

Table 6.32. Outcome distribution for pain and musculoskeletal symptoms

Pain and musculoskeletal symptom item	**Baseline mean ± SD**	**Post-intervention mean ± SD**	**Mean change**
I experience pain in my neck during or after tailoring activities.	3.66 ± 0.78	3.12 ± 0.92	-0.53
I experience pain in my shoulders during or after sewing activities.	3.48 ± 0.84	2.93 ± 1.00	-0.55
I experience pain in my arms or elbows while performing tailoring work.	3.47 ± 0.77	2.94 ± 0.96	-0.53

I experience pain or discomfort in my wrists and hands during repetitive sewing activities.	3.53 ± 0.76	3.02 ± 0.96	-0.51
My musculoskeletal pain increases after prolonged working hours.	3.52 ± 0.88	3.01 ± 0.96	-0.51
I experience muscular stiffness or discomfort after completing my daily tailoring activities.	3.53 ± 0.79	2.97 ± 0.95	-0.55

Table 6.32 shows the differences between pre and post scores of the individual items in the Pain and Musculoskeletal Symptoms questionnaire, with mean response scores decreasing for all six statements. The mean score for neck pain during or after tailoring activities dropped from 3.66 +- 0.78 at baseline to 3.12 +- 0.92 post-intervention (a reported mean decrease of 0.53 points). Likewise, the mean decrease in its shoulder pain was 0.55 points from 3.48 +- 0.84 to 2.93 +- 1.00, reflecting a favourable reduction in pain in the shoulder. They also reported a reduction in pain in the arms/elbows during tailoring activities from 3.47 +- 0.77 to 2.94 +- 0.96, a 0.53 point decrease. In addition, wrist and hand discomfort in repetitive sewing tasks decreased from 3.53 +- 0.76 to 3.02 +- 0.96 (a decrease of 0.51 points), and pain due to working long hours decreased from 3.52 +- 0.88 to 3.01 +- 0.96 (a decrease of 0.51 points). The mean score for post-work stiffness in the muscles became more favourable, from 3.53 +- 0.79 to 2.97 +- 0.95, a reduction of 0.55 points was reported. These results show the improvements in various anatomical areas, such as neck, shoulders, arms, elbows, wrists and hands, which are especially of relevance for the tailoring of workers due to the repetitive movements and prolonged working postures they are subjected to during their activities. The post intervention mean scores are lower for all six of the statements so decreased scores represent decreased agreement with statements involving pain and discomfort. In summary, Table 6.32 shows positive trends in all six items of pain and musculoskeletal symptoms, indicating that participants' self-reported musculoskeletal symptoms decreased after the intervention period.

Table 6.33. Statistical analysis of pain and musculoskeletal symptom scores by study group

Group	Baseline mean ± SD	Post-intervention mean ± SD	Mean change	Paired t	df	p-value
Experimental	21.19 ± 2.06	15.12 ± 2.07	-6.07	52.77	57	<0.001
Control	21.16 ± 1.91	20.86 ± 2.05	-0.29	3.08	57	0.003

Table 6.33 shows the change in total Pain and Musculoskeletal Symptoms scores in the experimental and control groups, showing that there was a significantly larger decrease in the Pain and Musculoskeletal Symptoms scores in the experimental group. The mean score for the experimental group was 21.19 +- 2.06 at baseline, and 15.12 +- 2.07 after the intervention, a mean difference of 6.07 points. A significant difference was found between baseline and post-intervention assessments using the paired-samples t-test ($t = 52.77$, $df = 57$, $p < 0.001$), suggesting that the home-based exercise programme was very effective in reducing the level of reported pain and musculoskeletal symptoms among participants. The change in the mean score of the control group was relatively small and did not differ from the one found in the experimental group: from 21.16 +- 1.91 at baseline to 20.86 +- 2.05 (mean change: 0.29). The difference was also statistically significant ($t = 3.08$, $df = 57$, $p = 0.003$) but the amount of change was much less than that seen in the experimental group. This comparison of the mean reductions suggests that the experimental group had a significant more decrease in musculoskeletal symptoms than the control group throughout the study. The total scores are lower with fewer reported symptoms and less musculoskeletal discomfort, as the statements in the questionnaire are negative. Overall, the pattern of change from this exercise study (Table 6.33) was more favourable in the experimental group, with the majority experiencing a highly significant decrease in pain and musculoskeletal symptom score, than the control group, which experienced a relatively small decrease.

Table 6.34. Outcome distribution for upper-limb functional limitations

Upper-limb functional limitation item	Baseline mean ± SD	Post-intervention mean ± SD	Mean change
I experience difficulty raising my arms above shoulder level.	3.64 ± 0.80	3.08 ± 0.91	-0.56
I experience difficulty reaching behind my head or back.	3.60 ± 0.78	3.10 ± 1.01	-0.50
I experience difficulty performing repetitive hand and finger movements.	3.60 ± 0.83	3.03 ± 0.96	-0.58
I experience difficulty holding and handling tailoring tools for prolonged periods.	3.47 ± 0.77	2.96 ± 0.93	-0.52
I experience difficulty maintaining my upper-limb position while operating a sewing machine.	3.49 ± 0.67	2.96 ± 0.91	-0.53
My upper-limb discomfort limits my ability to complete tailoring activities efficiently.	3.64 ± 0.83	3.10 ± 1.03	-0.53

Table 6.34 shows that there were decreases in mean scores at baseline and post-intervention for the items in the individual Upper-Limb Functional Limitations questionnaire, with favorable changes in the mean scores for the six statements, indicating reduction in the participants'

perceived ability to perform upper-limb activities related to tailoring work. In relation to difficulty raising arms above shoulder level, the mean score at baseline was 3.64 +- 0.80, while the post-intervention mean score was 3.08 +- 0.91, reducing the score by 0.56 points which implied that the participants' perceived ability to perform overhead movement improved. Likewise, it became more difficult to reach behind the head or back (3.60 +- 0.78 to 3.10 +- 1.01, reducing by 0.50 points). The mean scores of the participants, for repetitive movement of hands and fingers, fell from 3.60 +- 0.83 to 3.03 +- 0.96, reflecting a positive change in their ability to repeat manual tasks, which are integral with sewing and garment production tasks. In addition, the length of time using tailoring tools was reported as being difficult to hold and/or handle for an extended period decreased from 3.47 +- 0.77 to 2.96 +- 0.93, an improvement of 0.52 points. The challenges faced while using a sewing machine with the upper limbs also reduced from 3.49 +- 0.67 to 2.96 +- 0.91, and the difficulties in completing tailoring activities efficiently reduced from 3.64 +- 0.83 to 3.10 +- 1.03. The results of these findings showed good improvements on various aspects of upper-limb function, especially overhead reaching, repetitive hand activities, handling occupational tools and working positions. The negatively worded questionnaire statements result in lower scores at post intervention reflecting less agreement with statements that describe functional difficulties. In total, Table 6.34 shows that the participants' perceived functional ability of their upper limbs increased in all six functional limitation items, indicating positive changes after the study intervention.

Table 6.35. Statistical analysis of upper-limb functional limitation scores by study group

Group	**Baseline mean ± SD**	**Post-intervention mean ± SD**	**Mean change**	**Paired t**	**df**	**p-value**
Experimental	21.38 ± 1.80	15.33 ± 1.99	-6.05	51.96	57	<0.001

Control	21.52 ± 1.84	21.12 ± 1.97	-0.40	4.04	57	<0.001

Table 6.35 shows a significant difference between experimental and control group in total ULFL scores between baseline and post-test, with a greater decrease in ULFL scores in the experimental group. The mean score of the experimental group at baseline was 21.38 +- 1.80, while the mean score of the control group was 21.52 +- 1.84, which were relatively similar levels of perceived upper-limb functional limitations. After eight weeks, the mean score of the experimental group dropped drastically to 15.33 +- 1.99 (mean decrease of 6.05 points), and the paired-samples t-test showed a highly statistically significant difference between the pre- and post-intervention tests (with t = 51.96, df = 57, p less than 0.001). The control group, however, showed a small, but statistically significant, decrease in its mean score, from 21.52 +- 1.84 to 21.12 +- 1.97 (mean decrease of 0.40 points; t = 4.04, df = 57, p less than 0.001). The amount of change, however, was significantly higher in the experimental group than in the control group, suggesting a more positive perception of ability to perform upper limb activities in the experimental group. The total scores are lower scores as the questionnaire statements are negatively worded, indicating fewer functional difficulties with reaching, repetitive hand movements, using upper-limb tools and maintaining upper-limb positions during occupational activities. In general, Table 6.35 shows that the upper-limb functional limitation scores were significantly lower in the participants who received the home-based exercise programme, indicating that there was a positive relationship between the programme and the improvements in the participants' perceived occupational and daily functional abilities.

Table 6.36. Outcome distribution for work-related impact and quality of life

Work-related impact / quality-of-life item	**Baseline mean ± SD**	**Post-intervention mean ± SD**	**Mean change**

My musculoskeletal symptoms interfere with my daily household activities.	3.68 ± 0.79	3.14 ± 0.99	-0.54
My physical discomfort affects my ability to complete my regular working hours.	3.58 ± 0.85	3.03 ± 1.00	-0.55
My musculoskeletal symptoms affect my sleep and rest.	3.47 ± 0.88	2.96 ± 1.07	-0.52
My physical discomfort reduces my ability to participate in social and recreational activities.	3.59 ± 0.85	3.04 ± 1.02	-0.55
I feel physically exhausted after performing my regular tailoring activities.	3.47 ± 0.80	3.00 ± 1.00	-0.47
My musculoskeletal problems reduce my overall satisfaction with my physical health.	3.65 ± 0.85	3.12 ± 0.97	-0.53

Table 6.36 shows the differences between the Work-Related Impact and Quality of Life questionnaire items at baseline and post-intervention, highlighting how the mean response scores decreased in all six statements and indicating that there were reported favourable changes for the participants. Musculoskeletal symptoms interference with daily household activities mean score showed a decrease from 3.68 +- 0.79 at baseline to 3.14 +- 0.99 postintervention (0.54 points).

Likewise, the influence of physical discomfort on completing the regular working hours declined from 3.58 +/- 0.85 to 3.03 +/- 1.00, which represents a decrease of 0.55 points and indicates an improvement in participants' perceived ability to complete the occupational tasks. Good improvements were also reported for musculoskeletal symptoms that affect sleep and rest, with the mean score dropping from 3.47 +- 0.88 to 2.96 +- 1.07 (-0.52 point). Similarly, the effects of physical discomfort on participation in social and recreational activities decreased from 3.59 +- 0.85 to 3.04 +- 1.02, which suggests less interference with social participation. Physical exhaustion from completing daily tailoring activities reduced from 3.47 +- 0.80 to 3.00 +- 1.00 and dissatisfaction with overall physical health associated with musculoskeletal issues reduced from 3.65 +- 0.85 to 3.12 +- 0.97. The greatest reported mean change was for the impact on regular working hours and social and recreational participation, where both showed a mean change of around 0.55 points. The results indicate positive shifts in various aspects of daily and occupational functioning such as household duties, work hours, sleep, social involvement, and self-reported physical health. All six statements refer to negative consequences of musculoskeletal symptoms, so lower scores at post intervention reflect less agreement with negative consequences. Table 6.36 indicates that the overall trends for all six domains of Work-Related Impact and Quality of Life were positive, with no domain showing a worsening trend after the intervention period, suggesting that the worsening impacts of musculoskeletal problems in participants' daily and occupational activities were reduced.

Table 6.37. Statistical analysis of work-related impact and quality-of-life scores by study group

Group	Baseline mean ± SD	Post-intervention mean ± SD	Mean change	Paired t	df	p-value

Experimental	21.48 ± 1.88	15.53 ± 2.19	-5.95	47.96	57	<0.001
Control	21.41 ± 2.00	21.03 ± 2.04	-0.38	3.55	57	<0.001

Table 6.37 compares the changes in total Work-Related Impact and Quality-of-Life scores between the two groups (experimental group and control group) with paired-samples t-tests, showing a significantly higher decrease in Work-Related Impact scores of those who received the home-based exercise programme. Perceived work related impact and QOL problems were relatively similar across the two groups at the start of the study with the experimental group having a mean score of 21.48 +- 1.88 and the control group having a mean score of 21.41 +- 2.00. The mean score of the experimental group was 15.53 +- 2.19 after the intervention, with a mean change of 5.95 points and a paired-samples t-test that showed a highly statistically significant difference between pre- and post-intervention assessments (t = 47.96, df = 57, p less than 0.001). The mean score of the control group decreased, from 21.41 +- 2.00 to 21.03 +- 2.04, reflecting a mean change of -0.38 points, also statistically significant (t = 3.55, df = 57, p less than 0.001). Although the experimental group showed a statistically significant reduction, the magnitude of reduction was much greater in the experimental group, 5.95 points, than in the control group, 0.38 points. These results suggest that the perceived impact of musculoskeletal symptoms on household activities, working hours, sleep, social participation and on physical health was more positive among those who received the intervention. The statements in the questionnaire are presented in a negative direction and therefore lower total scores are indicative of less negative work-related impact and fewer reported difficulties in daily activities. In general, Table 6.37 shows a significant lower score in Work-Related Impact and Quality-of-Life scores in the experimental group, implying positive changes with the 8-week home-based exercise program.

Table 6.38. Weekly adherence to the home-based exercise programme

Study week	Participants assessed	Mean adherence (%)	SD	Median (%)
Week 1	58	87.1	6.8	87.0
Week 2	58	88.3	6.8	88.0
Week 3	58	86.1	7.0	87.0
Week 4	58	85.5	5.7	85.0
Week 5	58	86.3	7.2	86.0
Week 6	58	87.7	8.0	88.0
Week 7	58	87.6	7.5	88.5
Week 8	58	88.1	7.5	88.5

The weekly participation of the participants in the experimental group in the 8 week home based exercise programme is highlighted in Table 6.38 and shows that the participation was relatively consistent throughout the intervention period. All 58 participants were assessed during the first week and their level of adherence was found to be 87.1% +- 6.8%, and a median of 87.0%, which meant it was relatively high at the start of the programme. The mean adherence was 88.3% +- 6.8% (median 88.0%) in the second week, and reached the highest level during the 8-week period. In week three, mean adherence was 86.1% +- 7.0% and in the fourth week (lowest mean adherence for the programme), it was 85.5% +- 5.7%. However, compliance was greater than 85% during this time. During the 5th week, mean adherence was 86.3% +- 7.2% with a median of 86.0%, and then rose to 87.7% +- 8.0% in the 6th week with median 88.0%. In the seventh week, mean adherence was 87.6% +- 7.5% with a median of 88.5% and in the final week, mean adherence was 88.1% +- 7.5% with a median adherence of 88.5%. The standard deviations were between 5.7%

and 8.0% demonstrating that there was a bit of variability of adherence between participants during the intervention period. In addition, all 58 participants were evaluated weekly throughout the eight weeks of the programme indicating that there was consistent participant follow-up throughout the programme. The weekly mean adherence percentages were between 85.5% and 88.3%, indicating generally stable adherence levels with no significant drop-off towards the end of the intervention. Table 6.38 shows that overall there was a consistent pattern of adherence to the home based exercise programme over the course of the 8 week intervention, indicating sustained adherence to the prescribed exercises among participants. These results offer valuable background information for interpreting the changes in musculoskeletal symptoms, in functional limitations and ROM of the upper limbs.

DISCUSSION

The present study, "A Study to Evaluate the Effectiveness of a Home Based Exercise Program on Work Related Upper Limb Musculoskeletal disorders in Tailors - An Experimental Study" was conducted to evaluate the effectiveness of home based exercise program in reducing musculoskeletal symptoms and improving functional ability and range of motion of upper limb in tailors. 116 students were included in the study, 58 students in the experimental group and 58 students in the control group. Major outcome domains were pain and musculoskeletal symptoms, upper-limb functional limitations, work-related impact and quality of life and upper-limb range of motion. The results revealed a positive trend of improvement in the experimental group in the outcome measures.

Tailoring is a job which often includes long sitting, repetitive use of the upper limbs, continuous visual concentration, and neck flexion. Workers can experience significant biomechanical stress in the neck, shoulders, arms, elbows, wrists and hands from these job requirements. Banerjee et al. reported a musculoskeletal disorders prevalence of 65.45% among tailors of an urban community in Kolkata, with the highest prevalence being that of the neck (41.8%). They also found that working longer hours, having more professional experience and having workstations that were not ergonomically favourable were important occupational risk factors [1]. Likewise, a study of sewing machine operators in Bangladesh revealed the neck and back posture, repetition of upper limb movements and workstation design to be important ergonomic issues, with the lack of rest breaks being a significant concern [2]. This is evidence of the occupational relevance of the current study and provides a justification for the need for accessible interventions for musculoskeletal health among tailing workers.

7.1 Home-based exercise and musculoskeletal symptoms and pain

The first significant result of the current study was that the home-based exercise programme led to decreased pain and musculoskeletal symptoms after 8 weeks. This experimental group score

showed a mean decrease of 6.07 points from baseline to post-intervention (21.19 +- 2.06 to 15.12 +- 2.07) on the total Pain and Musculoskeletal Symptoms score. The control group, by comparison, showed a mean change of 0.29 points (from 21.16 +- 1.91 to 20.86 +- 2.05). A between group comparison also showed a significant mean difference change between the groups (-5.78), Welch's t (-38.67) and p less than 0.001.

In addition, the item-wise analysis revealed positive item changes in pain pertaining to the shoulder, arm, elbow, wrist, and hands, but not the neck. A decrease in discomfort around the shoulders and stiffness in the muscles after work was especially significant. These results indicate that the exercise programme might have helped participants become more muscually flexible and able to tolerate movement and become physically fit. Activities such as neck mobility, shoulder movements, scapular strengthening and wrist flexibility may have contributed to meeting physical demands in repetitive tailoring tasks.

These findings agree with those of Banerjee et al., which showed that musculoskeletal symptoms were a significant problem for the tailors and the need for better occupational posture and workstation ergonomic design was recommended [1]. Similarly, musculoskeletal pain was found to be reduced after an ergonomic educational intervention in tailors [3].

Jakobsen et al. examined the setting in which physical exercise is delivered (workplace or home based) to healthcare workers and found that the setting in which physical exercise is delivered may impact participation and intervention outcomes. They also showed that adherence is important in assessing the effectiveness of exercise programmes [4]. Even though the participants and exercise programs were different from those of the current study, their results can be used in support of the use of structured exercise in occupational musculoskeletal rehabilitation.

7.2 Upper-Limb Functional Limitations: Effect of Home-Based Exercise

Another significant finding was that upper limb functional ability had been improved as perceived by members of the experimental group. The mean pre- to post-increase for the total score for the Upper-Limb Functional Limitations score was 6.05 points, from 21.38 +- 1.80 at baseline to 15.33 +- 1.99 post-intervention. The control group, on the other hand, showed a decrease from 21.52 +- 1.84 to 21.12 +- 1.97, which equates to a mean decrease of just 0.40 points. Between group comparison showed statistically significant differences in changes (mean difference = -5.66, Welch's t = -37.12, $p<0.001$).

Individual questionnaire results also revealed improvements in overhead reaching, reaching behind head or back, repetitive hand and finger movements, using tailoring implements and upper limb position when using a sewing machine. These are functional activities that relate directly to the occupational requirements of tailoring workers. Better shoulder mobility, scapular control, muscular endurance and wrist flexibility may help to improve tolerance of repetitive occupational movements.

This improvement noted in the current study may relate to the variety of exercises (aerobic, resistance and flexibility) in the intervention protocol. Upper-limb activities included arm circles, shadow boxing, wall push-ups, scapular squeezes, shoulder external rotation exercises and wrist stretches targeting various aspects of upper-limb function.

Bates et al. found positive results of an upper-limb structured home exercise program on upper-limb function in a randomized controlled trial of older people in the community [5]. While the population was different, the study provides information that is relevant to supporting the use of home-based exercise to improve upper-limb function. The results of the current study also highlight the possible use of structured home-based exercises for workers who have repetitive upper-limb demands at work.

7.3 Work-Related Impact/Quality of Life

The current study also found positive effects on the work-related impact and perceived quality of life after the intervention. The total score for the experimental group dropped from baseline 21.48 +- 1.88 to post-intervention 15.53 +- 2.19, a mean of 5.95 points. The control group had a less reduction from 21.41 +- 2.00 to 21.03 +- 2.04, which is a mean change of 0.38 points. Between group comparison showed that the difference is statistically significant (mean difference = -5.57, Welch's t = -34.04, p less than 0.001).

The item-wise results showed positive changes in the interference of musculoskeletal symptoms in household activities, regular working hours, sleeping, social participation, physical exhaustion and perceived physical health. This is significant because musculoskeletal disorders in the workplace can impact not only physical performance, but also a person's ability to perform their daily tasks and engage in social life.

The observed changes indicate that the increase in physical comfort and the ability of the upper limbs may have resulted in a decrease in perceived interference with daily activities. However, the questionnaire items assessed work related impact and quality of life dimensions and did not assess work productivity, and did not use a standard generic quality of life measure. Thus the results need to be understood within the context of participants' experiences as reported.

Duffield et al. reviewed the broader impact of musculoskeletal conditions on health and multimorbidity and the implications for individual functioning and healthcare systems [6]. The current results further highlight the need to take a proactive and rehabilitative approach to musculoskeletal symptoms that includes work and everyday life.

7.4.1 Increase ROM of upper limbs

An important clinical finding was the increase in the recorded range of motion in the shoulder, elbow and wrist. Statistically significant pre-post differences were found in the experimental group for all 11 movements evaluated: shoulder flexibility, shoulder extension, shoulder abduction,

shoulder internal rotation, shoulder external rotation, elbow flexibility, elbow extension, wrist flexibility, wrist extension, radial deviation and ulnar deviation.

The experimental group showed a 8-9 degree improvement in the movements that they were evaluated on. For instance, the range of motion for shoulder flexion increased by 9.09deg, shoulder abduction by 8.46deg, elbow flexion by 8.93deg and wrist extension by 9.24deg. The control group, in contrast, showed generally smaller changes, and some of the movements, such as shoulder flexion and shoulder extension and wrist extension, were not statistically significant.

Based on these findings it can be suggested that the exercise programme may have been a factor in improving joint mobility and movement capacity. Dynamic arm movements, shoulder mobility exercises, neck stretches, and flexor/extensor stretches of the wrists may have contributed to the changes observed.

Grobler et al. studied work-posture changes of sewing-machine operators and showed the significance of occupational posture in musculoskeletal health [7]. Their results corroborate the importance of taking into account the physical condition and the ergonomics of the workplace when dealing with the musculoskeletal issues of sewing-machine operators.

Improvements in ROM should not be considered in isolation of pain, functional and job requirements, however. The use of both joint mobility measures and self-reported symptoms offer a more comprehensive picture of the changes found in the current study.

7.5 Comparison Between Experimental and Control Groups

Results from the experimental and control groups showed that in all three factors the experimental group consistently showed a greater improvement than the control group on the range of motion movements assessed. Statistically significant differences among the three factors on the questionnaires were also found ($p<0.001$) in the group x time interaction analyses.

The experimental group had larger decreases compared to the control group for pain and musculoskeletal symptoms, upper-limb function and work-related impact. Likewise, statistically significant differences were found in favour of the experimental group for each of the between-group range-of-motion comparisons that were selected.

Overall, these results indicate that the structured home-based exercise programme was related to improved outcomes compared to the control condition over the 8-week exercise period. The overall trend of the comparisons between different outcome domains supports the potential benefits of exercise-based rehabilitation for tailing workers.

7.6 Exercise Adherence and Reliability of the Questionnaire.

The experimental group had generally good compliance with the intervention during the eight weeks. The mean adherence for each week was between 85.5% and 88.3%, with the highest mean adherence being in the second week. There was no significant decrease in mean compliance at the end of the program. The results indicate that in general, the participants could be engaged in the prescribed home-based exercises.

The reliability analysis of the questionnaire showed acceptable Cronbach's alpha values: 0.735 for Pain and Musculoskeletal Symptoms, 0.834 for Upper-Limb Functional Limitations and 0.849 for Work-Related Impact and Quality of Life. These values suggest acceptable to good internal consistency between the three six item factors.

In addition, high positive correlation was detected between the changes in the three factors of the questionnaire, Pearson correlation coefficient between 0.925 and 0.928 ($p < 0.001$). This shows that the changes in pain were strongly correlated with functional limitations and work-related impact. These correlations do not provide causal relationships between the outcome domains, however.

In summary, the results indicate that the home-based exercise programme was related to positive modification of musculoskeletal symptoms and perceived upper-limb function and joint mobility with relatively stable adherence during the intervention.

LIMITATIONS OF THE STUDY

Despite the positive findings of the present study after 8-weeks home-based exercise training, there are some limitations that should be taken into consideration to assess the findings and apply them to the broader tailoring population.

- Small sample size: Study sample size was 116 with 58 in each group. This sample allowed for comparison of the experimental and control groups, but additional subjects sampled from a variety of occupational settings would increase the generalizability and accuracy of the results.
- Limited geographical representation: The study was done with the tailoring workers of Skill Development Institute, Bhubaneswar, Odisha. Hence, the results do not necessarily reflect the total population of the tailoring workers especially those who work in other geographical areas, informal sectors, and home-based tailoring shops and other garment industries.
- Convenience sampling: Convenience sampling was used for recruitment of the participants which might have led to selection bias. The accessible or willing to participate individuals may have been different from those who were not; thus, the representativeness of the sample may be affected.
- Short duration of intervention: The intervention lasted for 8 weeks. Though improvements were noted over this study, it was not able to determine whether such improvements would continue over longer periods of time. It therefore remains to be seen whether continuing exercise participation will have long-term effects.
- Self-reported dependence: Questionnaire responses were used to assess pain, functional limitations and work-related impact. Self-reported measurements might be subject to individual perceptions and recall problems, and to different interpretations of the statements in questionnaires. Therefore, subjective changes do not necessarily reflect objective changes of physical performance.

- Researcher-developed questionnaire: These three questionnaire factors included researcher-developed items to assess Pain and Musculoskeletal Symptoms, Upper-Limb Functional Limitations, and Work-Related Impact and Quality of Life. The internal consistency analysis revealed acceptable to good Cronbach's alpha values but further validation is needed such as construct validity, test–retest reliability, and criterion validity before the questionnaire can be deemed a standardized measurement instrument.
- Occupational, ergonomic differences: Participants were different on tailoring activities, professional experience, working hours, sewing-machine types and rest-break patterns. These occupational factors may impact musculoskeletal symptoms and intervention responses. Subgroup analyses was conducted but the study might not have adequately included all possible ergonomic and occupational factors.
- Adherence to exercise: Weekly mean adherence was relatively high, but it is possible that each person in the study may have varied in the frequency, intensity and accuracy of the prescribed exercises. The home-based exercise programmes are flexible and may not provide as much direct supervision as supervised rehabilitation programmes.
- Limited work productivity assessment: Question statements about working time, household activities, sleep, social participation and physical exhaustion, were used to assess the work-related impact. There was no independent measurement of objective data like work output, absenteeism, time to complete task, and occupational efficiency.
- Limited follow-up assessment: There was follow-up assessment at baseline and at post-intervention, but there was no longer follow-up period. Thus, improvements and recurrence of musculoskeletal symptoms could not be assessed as to its sustainability.
- Potential for external factors: During the study period, factors such as workload, work station set-up, physical activity, and other health-related behaviors may have had an impact

on participants' outcomes. A full control of these factors may not have been possible during the intervention.

- Limited clinical assessment: Range of motion was assessed in 11 upper-limb movements, but other objective measures of upper-limb function, such as muscle strength, endurance, grip strength, scapular control, and standardized disability measurements, could have been included for a more comprehensive clinical evaluation.

Although the study has these limitations, it offers valuable data on the feasibility of using a structured home-based exercise programme for upper-limb musculoskeletal symptoms of work-related problems among tailoring workers. The results need to be interpreted in the light of the study population, the length of exposure to the intervention, the way the results were measured, and the statistical design of the study.

FUTURE RECOMMENDATIONS

The following recommendations are made for future research and the better management of work-related upper-limb musculoskeletal disorders in tailoring workers based on the findings and the limitations of the present study.

- Use larger samples in future studies: Future studies should involve larger samples drawn from various tailoring establishments, the garment industry and informal occupational settings. This would increase the representativeness of the participants and the generalizability of the findings.
- Add several locations: Other studies should be done in other cities and states to see if there are any differences in effectiveness of home-based exercise programmes depending on occupational conditions, workplace environments and demographic characteristics.
- Include longer intervention and follow up periods: Future studies should look at the effectiveness of home-based exercise programmes with a longer follow up period and/or intervention period of up to 12 or 16 weeks, as well as 3 and 6 months follow up. This would assist in determining if the improvement in pain, function and joint mobility are sustained.
- Include use of validated and standardized outcome measurement tools: Future studies should use standardized and validated tools for symptom assessment (e.g., Nordic Musculoskeletal Questionnaire) and upper limb disability (e.g., Disabilities of the Arm, Shoulder and Hand (DASH) questionnaire). Additionally, validated pain-intensity measures and suitable quality-of-life instruments should be taken into account.
- Implement overall ergonomic interventions: Exercise programmes should be accompanied by ergonomic training that includes proper sitting posture, workstation height, chair support, position of the sewing machine, lighting and regular rest breaks. This can be done for both physical fitness and work risk factors.

- Objectively assess functional outcomes: Objective measures (such as grip strength, muscle endurance, functional reach, upper-limb task performance, and standardized joint range-of-motion measures) should be included in future studies. These assessments would be used to gain more information than just self reported improvements.
- Create tailored exercise regimes: Exercise programming should take into account participants' age, occupational exposure, severity of symptoms, and physical abilities and functional demands. Individualized progression can enhance exercise tolerance and adherence and safety.
- Different methods for delivering exercise: Future studies need to be conducted to compare home-based exercise, workplace-based exercise, and physiotherapy supervised exercise programmes to determine which is the most effective and feasible way of tailoring workers to their exercise needs. Additionally, the role of direct supervision and educational support on adherence should be explored.
- Enforce adherence monitoring: Digital exercise diaries and mobile apps, video instructions and regular physiotherapist follow-up could be helpful. The next step is to evaluate whether these approaches are effective in increasing adherence and long-term engagement.
- Cost-effectiveness evaluation: Future studies should address the financial viability of home-based exercise programmes, including the costs of the intervention, health-care utilization and work absenteeism and improvements in occupational functioning.
- Explore gender and occupational differences: Further research should be conducted to determine if there are gender or occupational differences in response to interventions. Subgroup analyses might be adequate with larger samples.
- Well-designed multicentre randomized controlled trials, using appropriate allocation concealment, standardizing the intervention and assessing outcomes blind, if possible, and

applying an intention-to-treat analysis, would provide stronger evidence of the effectiveness of the interventions.

- Embed prevention and education: There should be a further focus on musculoskeletal symptom recognition, safe working practice, taking movement breaks, safe progression of exercises and awareness of risk factors for musculoskeletal conditions in future programmes.

A more holistic strategy to deal with WMSDs with tailoring workers is the incorporation of exercise therapy and ergonomic modification and occupational health education. Further studies should not only aim to alleviate symptoms in the short-term, but to preserve functional independence and to enhance occupational participation, and to prevent return of musculoskeletal complaints.

CONCLUSION

This study aimed to assess the efficacy of an 8-week home-based exercise program for reducing upper limb musculoskeletal disorders (UL-MDS) in tailoring workers. The study was based on three major dimensions of musculoskeletal health: pain and musculoskeletal symptoms, upper-limb functional limitations and work-related impact and quality of life. Furthermore, ROM of the shoulder, elbow and wrist was measured to assess changes in upper-limb mobility.

The results showed a good trend in improvement for the experimental group who completed the home exercise program. There was a large decrease in Pain and Musculoskeletal Symptoms score, with a mean decrease of 6.07 points in the experimental group versus 0.29 points in the control group. Likewise, the Upper-Limb Functional Limitations scores of the experimental group fell 6.05 points while the control group fell 0.40 points. A significantly larger decrease was also found in the experimental group compared to the control group for Work-Related Impact and Quality of Life scores (5.95 points and 0.38 points, respectively). The results of the between group comparisons showed significant difference for all three factors of the questionnaire ($p<0.001$).

This study also showed that the ROM of the upper-limb was improved for all the movements evaluated at the shoulder, elbow and wrist. Statistically significant pre-post changes were observed in all 11 movements in the experimental group between shoulder flexion, extension, abduction and internal and external rotation at the shoulder, elbow flexion and extension, and wrist flexion, extension, radial deviation and ulnar deviation. The results showed positive improvements in joint mobility after the exercise programme.

Additionally, there was good adherence across the 8-week intervention in the experimental group, with the weekly mean adherence ranging from 85.5% to 88.3% across the 8 weeks. The reliability of the questionnaire was also analyzed, which showed that the internal consistency of the questionnaire items on the three factors was acceptable to good, with Cronbach's Alpha ranging from 0.735 to 0.849.

The results should all together indicate that a well organised home training programme of aerobic, resistance and flexibility exercises could be beneficial for the management of the upper-limb musculoskeletal symptoms of tailors working at their workstations. It is a potentially accessible approach to rehabilitation because of the relatively simple nature of the exercises, the lack of equipment needs, and the flexibility in delivery.

The study also highlights the need for the integration of exercise-based rehabilitation, ergonomic awareness and optimal occupational practice. An increased level of physical activity, appropriate working postures and adequate rest breaks, combined with early identification and prevention of musculoskeletal symptoms, should be encouraged to ensure the long term musculoskeletal health of tailoring workers.

To conclude, the present study justifies the possible effectiveness of an eight-week home exercise programme in decreasing SBP, FL and improving upper limb range of motion among tailing workers. This is reflected in the better improvements seen within the experimental group, as compared to the control group, and suggests a positive response to the intervention. However, additional research with larger sample sizes and longer follow-up is recommended to provide information on the long-term effectiveness and generalizability of this intervention.

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ABSMARI ETHICS COMMITTEE

ABHINAV BINDRA SPORTS MEDICINE AND RESEARCH INSTITUTE, BHUBANESWAR, ODISHA

CDSCO Reg. No.: ECR/1981/Inst/OD/24

Prof. (Dr.) E. Venkata Rao
Chairperson

Mr. Chinmaya Kumar Patra
Member Secretary

Ref. No. ABSMARI/IEC/2025/308

Date: 05/02/2026

APPROVAL LETTER
APPENDIX- VIII

To,

ANKITA SAHAI
ABSMARI
273, PAHAL, BHUBANEWAR-752101

MEMBERS

Dr. Smaraki Mohanty
Clinician

Dr. Satyajit Mohanty
Scientific Member

Mr. Shib Shankar Mohanty
Legal Expert

Ms. Annie Hans
Social Scientist

Ms. Subhashree Samal
Lay Person

Mr. Deepak Ku. Pradhan
Scientific Member

IEC-SECRETARIAT

Mr. Gouranga Ku. Padhy
Mr. Susant Ku. Raychudamani

Protocol Title: A Study to Evaluate the Effectiveness of A Home-Based Exercise Program on work-related Upper-Limb Musculoskeletal Disorders among tailors- An Experimental Study.

Protocol ID.: ABS-IEC-2025-PHY-100

Subject: Approval for the conduct of the above referenced study

Dear Mr./Ms./Dr **ANKITA SAHAI**
With reference to your Submission letter dated 10/10/2025 the ABSMARI IEC has reviewed and discussed your application for conduct of the study on dated 15/10/2025.

The following documents were reviewed and discussed

S.N.	Documents	Document (Version/Date)
1	IEC Application Form	15/10/2025
2	Informed Consent Form	15/10/2025
3	Undertaking form PI	15/10/2025
4	CRF	15/10/2025
5	COI from the Investigators	15/10/2025

The following members were present at meeting held on 15/10/2025

ABSMARI ETHICS COMMITTEE * BHUBANESWAR * Dt.....

1

Utkal Signature, Plot No.-273, Ground Floor, Pahal, Bhubaneswar-752101
+91-63707-03654
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ABSMARI ETHICS COMMITTEE

ABHINAV BINDRA SPORTS MEDICINE AND RESEARCH INSTITUTE, BHUBANESWAR, ODISHA

CDSCO Reg. No.: ECR/1981/Inst/OD/24

Prof. (Dr.) E. Venkata Rao
Chairperson

Mr. Chinmaya Kumar Patra
Member Secretary

Ref. No. ABSMARI/IEC/2025/308

Date: 05/02/2026

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Mr. Deepak Ku. Pradhan
Scientific Member

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Mr. Gouranga Ku. Padhy
Mr. Susant Ku. Raychudamani

S.N.	Name of the Member	Designation & Qualification	Representation as per NDCT 2019	Gender (M/F)	Affiliation with the Institution (Y/N)
1	Prof. Dr. E. Venkata Rao	Professor (MBBS, MD, Dept. of Community Med.) IMS & Sum Hospital, BBSR	Chair Person	M	N
2	Dr. Smaraki Mohanty	Asst. Prof-IMS & Sum Hospital/MBBS, MD (Community Med)	Clinician	F	N
3	Mr. Shiba Sankar Mohanty	Junior Counsel-Lt. Ramachandra Sarangi's Chamber / BA LLB	Legal Expert	M	N
4	Mr. Chinmaya Kumar Patra	Principal-ABSMARI, MPT	Member Secretary	M	Y
5	Ms. Subhashree Samal	Ret. Reader-Pol Sc.	Lay Person	F	N
6	Mr. Deepak Kumar Pradhan	Asst. Prof-ABSMARI, MPT	Scientific Member	M	Y
7	Dr. Satyajit Mohanty	Director-Medcare Hospital, BBSR	Scientific Member	M	N

This is to confirm that only members who are independent of the Investigator and the Sponsor of the trial have voted/ provided opinion on the trial.

This Committee approves the documents and the conduct for the study in the presented form with necessary recommendation.

The ABSMARI IEC must be informed about the progress of the study in the prescribed format attached, any SAE occurring in the course of the study, any changes in the protocol and patient information/informed consent/assent and request to provide a copy of the final report.

The ABSMARI IEC follows procedures that are in compliance with the requirements of ICH (International Conference on Harmonization) guidance related to GCP (Good Clinical Practice) and applicable Indian regulations.

Yours sincerely,

Mr. Chinmaya Kumar Patra
Member Secretary
ABSMARI Ethics Committee
Pahal, Bhubaneswar

Member Secretary
ABSMARI ETHICS COMMITTEE

ABSMARI ETHICS COMMITTEE Dt..... BHUBANESWAR

2

Utkal Signature, Plot No.-273, Ground Floor, Pahal, Bhubaneswar-752101 +91-63707-03654 iec@absmari.com

Informed Consent form to participate in a clinical trial

Study Title: A STUDY TO EVALUATE THE EFFECTIVENESS OF A HOME-BASED EXERCISE PROGRAM ON WORK-RELATED UPPER-LIMB MUSCULOSKELETAL DISORDERS AMONG TAILORS- AN EXPERIMENTAL STUDY.

Study Number:

Subject 's Name: Avadh sankar Subject 's Initials: _____

Date of Birth / Age: _____

Address of the Subject_____

Qualification _____

Occupation: Student/Self-Employed/ Service/Housewife/Others (Please tick as appropriate)

Annual Income of the subject _____ if applicable

Name and address of the nominee(s) and his relation to the subject

_____ (for the purpose of compensation in case of trial related death).]

(i) I confirm that I have read and understood the information sheet dated _____ for the above study and have had the opportunity to ask questions.

(ii) I understand that my participation in the study is voluntary and that I am free to withdraw at any time, without giving any reason, without my medical care or legal rights being affected.

(iii) I understand that the Sponsor of the clinical trial, others working on the Sponsor 's behalf, the Ethics Committee and the regulatory authorities will not need my permission to look at my health records both in respect of the current study and any further research that may be conducted in relation to it, even if I withdraw from the trial. I agree to this access. However, I understand that my identity will not be revealed in any information released to third parties or published.

(iv) I agree not to restrict the use of any data or results that arise from this study provided such a use is only for scientific purpose(s)

(v) I agree to take part in the above study.

Signature (or Thumb impression) of the Subject/Legally Acceptable Representative:

_____

Date: ____/ ____/ ____

Signatory 's Name: _____

Signature of the Investigator: _____

Date: Study Investigator 's Name: _____

Signature of the Witness: _____

Date: _____/ _____/ _____

Name of the Witness: _____

*Copy of the Patient Information Sheet and duly filled Informed Consent Form shall be handled over to the subject or his/her attendant.

ANNEXURE

STRUCTURED RESEARCH QUESTIONNAIRE AND CLINICAL ASSESSMENT FORM

For MPT Research Study

Research Title: A Study to Evaluate the Effectiveness of a Home-Based Exercise Program on Work-Related Upper-Limb Musculoskeletal Disorders Among Tailors – An Experimental Study.

Study Setting: Skill Development Institute, Bhubaneswar, Odisha.

Participant information	**Details**
Participant Identification Number	_____
Date of Assessment	_____
Group	Experimental / Control
Assessment	Baseline / Post-intervention

SECTION A: DEMOGRAPHIC AND OCCUPATIONAL INFORMATION

Instructions: Please select or fill in the appropriate response.

No.	**Demographic / occupational variable**	**Response options**
1	1. Age (completed years)	20–25 / 26–30 / 31–35 / 36–40 / 41–45
2	2. Gender	Male / Female / Other
3	3. Marital status	Married / Unmarried / Other
4	4. Educational qualification	No formal education / Primary / Secondary / Higher secondary / Graduate and above
5	5. Type of tailoring work	Sewing-machine operator / Independent tailor / Garment stitching / Cutting and stitching / Other: _____
6	6. Total years of professional tailoring experience	1–5 / 6–10 / 11–15 / 16–20 / Above 20 years
7	7. Average working hours per day	Below 4 / 4–6 / 7–8 / 9–10 / Above 10 hours
8	8. Number of working days per week	1–3 / 4–5 / 6 / 7 days

9	9. Frequency of rest breaks	Every 30 minutes / Every hour / Every 2 hours / Occasionally / No regular breaks
10	10. Type of sewing machine mainly used	Manual / Electric / Industrial / Overlock / Other: _____

SECTION B: FACTOR 1 – PAIN AND MUSCULOSKELETAL SYMPTOMS

Objective: To assess changes in work-related upper-limb pain and musculoskeletal symptoms among tailors following the home-based exercise programme.

RESPONSE SCALE AND SCORING

Please select one response for each statement in Sections B, C and D.

Response	**Score**
Strongly disagree	1
Disagree	2
Neutral	3
Agree	4
Strongly agree	5

Outcome measure: Pain intensity and musculoskeletal symptoms.

Item code	**Statement**	**Response (1–5)**
P1	I experience pain in my neck during or after tailoring activities.	_____
P2	I experience pain in my shoulders during or after sewing activities.	_____
P3	I experience pain in my arms or elbows while performing tailoring work.	_____
P4	I experience pain or discomfort in my wrists and hands during repetitive sewing activities.	_____
P5	My musculoskeletal pain increases after prolonged working hours.	_____
P6	I experience muscular stiffness or discomfort after completing my daily tailoring activities.	_____

SECTION C: FACTOR 2 – UPPER-LIMB FUNCTIONAL LIMITATIONS

Objective: To evaluate changes in upper-limb functional ability following the eight-week home-based exercise programme.

Outcome measure: Functional ability and occupational performance.

Item code	Statement	Response (1–5)
F1	I experience difficulty raising my arms above shoulder level.	_____
F2	I experience difficulty reaching behind my head or back.	_____
F3	I experience difficulty performing repetitive hand and finger movements.	_____
F4	I experience difficulty holding and handling tailoring tools for prolonged periods.	_____
F5	I experience difficulty maintaining my upper-limb position while operating a sewing machine.	_____
F6	My upper-limb discomfort limits my ability to complete tailoring activities efficiently.	_____

SECTION D: FACTOR 3 – WORK-RELATED IMPACT AND QUALITY OF LIFE

Objective: To assess the impact of work-related musculoskeletal symptoms on daily activities, occupational participation and quality of life.

Outcome measure: Self-reported quality of life and daily functional impact.

Item code	Statement	Response (1–5)
Q1	My musculoskeletal symptoms interfere with my daily household activities.	_____
Q2	My physical discomfort affects my ability to complete my regular working hours.	_____
Q3	My musculoskeletal symptoms affect my sleep and rest.	_____
Q4	My physical discomfort reduces my ability to participate in social and recreational activities.	_____
Q5	I feel physically exhausted after performing my regular tailoring activities.	_____
Q6	My musculoskeletal problems reduce my overall satisfaction with my physical health.	_____

SECTION E: CLINICAL RANGE OF MOTION (ROM) ASSESSMENT

Objective: To evaluate changes in the range of motion of affected upper-limb joints following the exercise programme.

Measurement instrument: Inclinometer. Assessment to be performed by the physiotherapist/researcher. Unit: degrees (°).

Joint	Movement	Right side (°)	Left side (°)
Shoulder	Flexion		
Shoulder	Extension		
Shoulder	Abduction		
Shoulder	External rotation		
Shoulder	Internal rotation		
Elbow	Flexion		
Elbow	Extension		
Wrist	Flexion		
Wrist	Extension		
Wrist	Radial deviation		
Wrist	Ulnar deviation		

Pre- and post-intervention comparison:

Joint movement	Baseline	Post-intervention	Difference
Shoulder flexion			
Shoulder abduction			
Shoulder external rotation			
Elbow flexion			
Wrist extension			

Record movements on both sides where relevant. Use a standardized measurement position and the same assessment procedure at baseline and follow-up.

SECTION F: EXERCISE ADHERENCE MONITORING

To be completed during the eight-week intervention by participants in the experimental group.

Week	Exercises completed regularly?	Difficulty experienced	Pain/discomfort	Remarks
Week 1	Yes / No			
Week 2	Yes / No			
Week 3	Yes / No			
Week 4	Yes / No			
Week 5	Yes / No			
Week 6	Yes / No			
Week 7	Yes / No			
Week 8	Yes / No			

Reasons for missed exercise sessions (if any):

☐ Lack of time

☐ Pain or discomfort

☐ Work-related responsibilities

☐ Illness

☐ Lack of motivation

☐ Other: _____

SECTION G: FINAL CLINICAL ASSESSMENT RECORD

Outcome measure	Baseline assessment	Post-intervention assessment
Musculoskeletal symptom distribution		
Pain intensity (NPRS), if approved		
Functional ability score		

Shoulder ROM		
Elbow ROM		
Wrist ROM		
Quality-of-life score		